

AC SERVO DRIVE

TURBO *SD Series with* Ether**CAT** *Interface*

ver 1.0 (2016.1)

User Manual



Safety Precautions

I. Definition of Warning Symbols

WARNING	Caution
	

- Warning: Indicates a hazardous situation that could result in electric shock if not avoided.
- Caution: Indicates a hazardous condition or situation where failure to exercise caution may result in minor or serious injury or property damage.

II. Warning

- Do not use in locations containing corrosive or explosive gases.
- When using in areas where the servo motor is rotating or where machinery may cause hazardous situations, implement protective measures.
- To prevent electric shock, disconnect the power supply when installing, removing, or performing wiring work.
- To prevent electric shock and malfunction, ground the PE terminal on the L1(R), L2(S), L3(T) main power input side of the servo drive to a single point using a Type 3 grounding method (100Ω or less, Ø 1.6mm or larger). Specifically, use a wire at least 40 mm thicker than the power lines for the servo drive's FG terminal.
- To prevent electric shock, the ground terminal (FG) of the servo motor must be connected to the FG terminal of the servo drive's motor output connector.
- Ensure no hazards occur during servo motor installation, disassembly, removal, or relocation.
- After the servo motor stops during an emergency stop, disconnect the power supply (L1, L2, L3) to the servo drive.

III. Caution

- To prevent burns, do not touch the servo motor, the heat sinks of the servo motor or servo drive, or the regenerative resistor during operation or immediately after power is disconnected. Implement protective measures in areas where exposure is possible.
- Do not use the servo motor or servo drive under the following conditions, as they may be damaged:
 - Do not connect power to the U, V, W terminals of the servo motor.
 - Do not subject the servo motor to impact. This may damage the internal encoder.
 - Do not connect power to the servo drive's U, V, W terminals.
 - When performing withstand voltage tests or insulation tests, disconnect all connectors from the servo drive and ensure no test voltage is applied to the servo drive. Specifically, ensure no test

voltage is applied to the encoder connector terminals of the servo motor.

- Use servo motors and servo drives only in combinations with specified ratings.
- Prevent water or oil from directly contacting the servo motor. Use the servo motor in locations where water and oil cannot reach its lead wires.
- Do not use outside the specified operating conditions.

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Chapter 1

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1.1 Unpacking Verification

After receiving the AC servo drive, please ensure the following items are checked.

- 1) Verify that the product matches the ordered capacity.

Verify that the part number on the nameplate matches the ordered part number.

(The capacity is indicated within the part number. For detailed explanation, refer to Section 1.2.)

- 2) Check for any damage.

Inspect the device to ensure it was not damaged during shipping.

- 3) Check if the case or screws are loose.

Verify that the shielding on the exterior case is intact and that all essential screws are securely tightened. If any items are damaged or mismatched, please notify the sales outlet where you purchased the product or our headquarters.

- 4) A complete and functional AC servo system includes the following components.

Part I: Standard Supplies

- (1) Servo Drive
- (2) 11-pin terminal block (for L1C, L2C, PI, PO, B1+, B2, B3, L1, L2, L3, EE)
- (3) 3-pin terminal block (for U, V, W)
- (4) Two jumper pins (for Po, PI connection and regenerative resistor connection)
- (5) User Manual

Part II: Optional Specification Parts

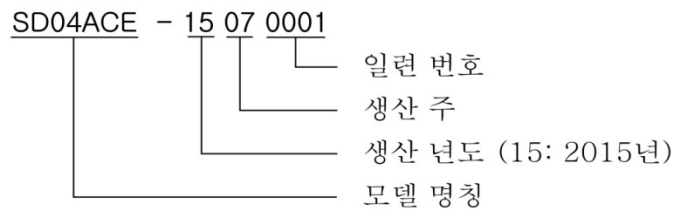
- (1) 1 power cable, for connecting the servo motor and servo drive, a 4P cable including a green ground wire. Connect the green ground cable to the ground terminal of the servo drive.
- (2) 1 Serial Encoder Connector, used when connecting to the servo motor's encoder terminal (CN2).
- (3) I/O Input Connector x1. Used when connecting to the external Input/Output terminals (CN1).
- (4) One EtherCAT input connector. Use when connecting to the EtherCAT Input/Output terminals (CN4/CN5).

1.2 Model Nameplate Description

1.2.1 Nameplate Description

모델 명칭	→	CSCAM AC Servo Drive
용량 사양	→	Model : TURBO SD-04ACE
입력 전력	→	Power : 400W
정격 출력 전류	→	Input : AC 200~230[V] 50/60Hz
일련 번호	→	Output : 3[A]
		Serial No : SD04ACE-15070001

1.2.2 Serial Number Description



1.3 Model Name Description

1.3.1 Servo Drive Model Information

TURBO SD	04	A	C	A
시리즈명	용량	입력전압	엔코더	지령방식
서보 시리즈	04 : 400W 10 : 1.0KW 13 : 13KW 20 : 2.0KW 30 : 3.0KW 50 : 5.0KW 75 : 7.5KW 110 : 11KW	A : 200V B : 400V	A: Quqdrature B: 17bit Serial D: 19bit Serial	A : Analog/Pulse E : EtherCAT C: 위치결정형

1.3.2 Servo Motor Model Information

TURBO SMG	04	A	A	A	1	1
시리즈명	용량	입력전압	엔코더	설계사양	축단사양	브레이크사양
SMG : 고 토크용 SMA : 고 회전용	01 : 100W 02 : 200W 04 : 400W 05 : 500W 08 : 750W 13 : 1.3KW 20 : 2.0KW 30 : 3.0KW 44 : 4.4KW 55 : 5.5KW 75 : 7.5KW	A : 200V B : 400V	A : Quqdrature B: 17bit Serial (중분치) C: 17bit Serial (절대치) D: 19bit Serial (중분치) E: 19bit Serial (절대치)	A : 표준 E : 고정도	1 : straight 2 : straight/key	1 : DC24 2 : DC90

1.4 Servo Drive / Motor Specifications

The table below shows the possible combinations of servo drives (SD series) and servo motors (SMG/SMA series).

Servo Pack Type SD-	04A	08A	10A	20A	30A	55A	75A	110A
Maximum Applicable Motor [KW]	0.4	0.75	1.0	2.0	3.0	5.5	7.5	11
Rated Output Current [A]	3.0	5.0	6.5	13.5	16.5	35.0	40.0	50.0
Maximum Output Current [A]	9.0	15.0	19.5	40.5	49.5	96.0	104.0	125.0
Recovery Resistor	Internal / External					External		
Main Circuit [3-phase: AC200V]	AC200 ~ 230V +10~-15% 50/60Hz							
Control Circuit [Single-phase: AC200V]	AC200 ~ 230V +10~-15% 50/60Hz							

Voltage		220V												
Flange Specifications		40	60		80		130			180				
Drive Format	SD-	04A	04A	04A	10A	10A	10A	20A	20A	20A	30A	55A	75A	110A
Motor Type	SMG-SMA-	01A□A	02A□A	04A□A	04A□SA	08A□A	05A□A	13A□A	20A□SA	20A□A	30A□A	44A□A	55A□A	75A□A
Rated Output	kW	0.1	0.2	0.4	0.4	0.75	0.45	1.3	2	1.8	2.9	4.4	5.5	7.5
Rated Torque	N·m	0.312	0.637	1.27	0.91	2.39	2.83	8.3	11.3	11.3	18.4	28.2	34.8	47.8
Peak Torque Torque	N·m	0.1	2.23	4.46	6.69	7.16	8.7	23.0	27.6	28.4	44.7	71.0	87.3	117
Rated Current	Arms	0.84	1.6	2.7	4.2	4.4	3.5	10.4	16	16.5	23.6	32.4	41.8	54.2
Instantaneous Maximum Current	Arms	2.9	5.8	9.3	14.9	13.4	10.5	27	42	41	54	83	108	129
Rated Speed Speed	min ⁻¹	3000					1500							
Maximum Rotation Speed	min ⁻¹	5000					3000							
Torque Constant	N·m /Arms	0.352	0.435	0.512	0.505	0.59	0.83	0.85	0.68	0.70	0.82	0.86	0.80	0.90
Rotor Moment of inertia ⁻¹	kg ^{-m²} 10 ⁻⁴	0.0525	0.259	0.442	0.667	0.672	7.22	20.3	26.7	31.4	45.8	67.3	88.7	126
		(0.074)	(0.323)	(0.506)	(0.744)	(0.86)	(9.3)	(22.4)	(34.2)	(40.0)	(54.3)	(75.7)	(97.3)	(133.2)
Rated Power Rate	kW/s	15.2	15.7	36.5	54.7	84.8	11.0	33.5	36.1	41.3	75.1	118	135	183

*1) The values in parentheses indicate values for motors equipped with a holding brake.

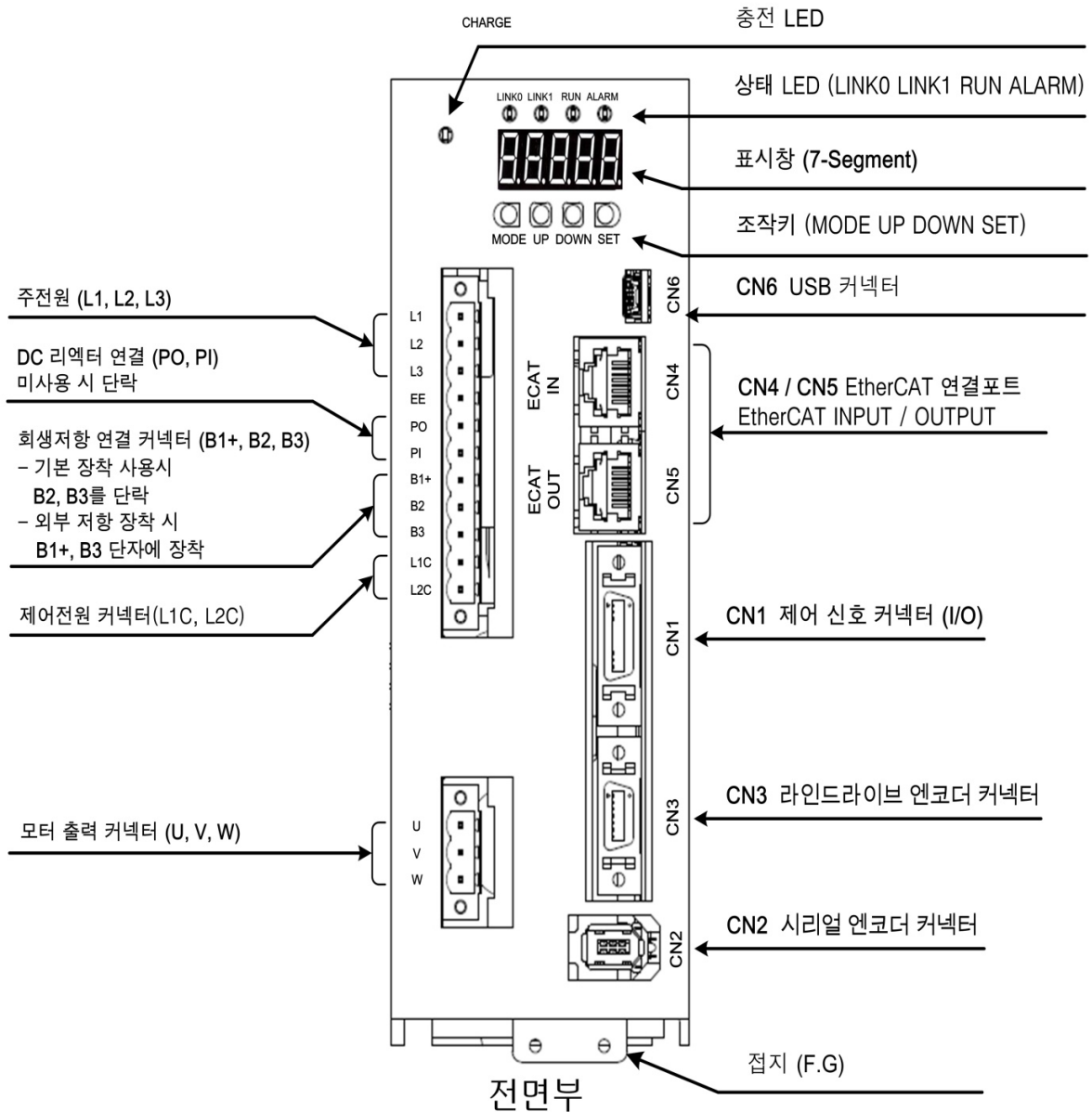
The servo drives listed in the table above are designed for use in combination with specific servo motors. Verify the specifications of the drive and motor you intend to use.

Ensure both the servo drive and motor have correctly matched capacity (rated power). If the power of the motor and drive is outside the specified range, the drive and motor may overheat and a servo alarm may occur.

The drives listed in the table above are designed based on the motor's rated current.

1.5 Servo Drive Structure and Description

1.5.1 Front Panel Structure and Description



Charging LED: The charging LED illuminates when the drive's main power supply voltage is present. Do not touch or disassemble the drive while it is illuminated.

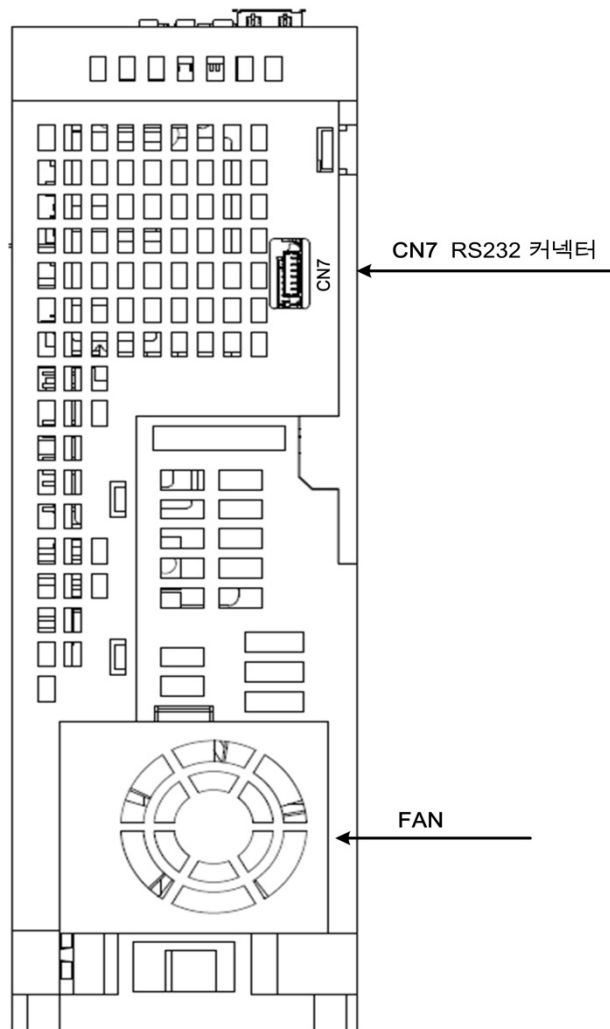
Status LED: Indicates the connection status and operating status of the EtherCAT port and displays alarm information.

Ground : Ensure the grounding of the motor, servo drive, and the control panel where the drive is installed is properly connected.

Operation Keys: Use the operation keys to set the parameters of the drive and motor.

Display Window: Shows the drive's operating status and allows parameter verification using the operation keys

1.5.2 Substructure and Description



Chapter 2

Product Specifications

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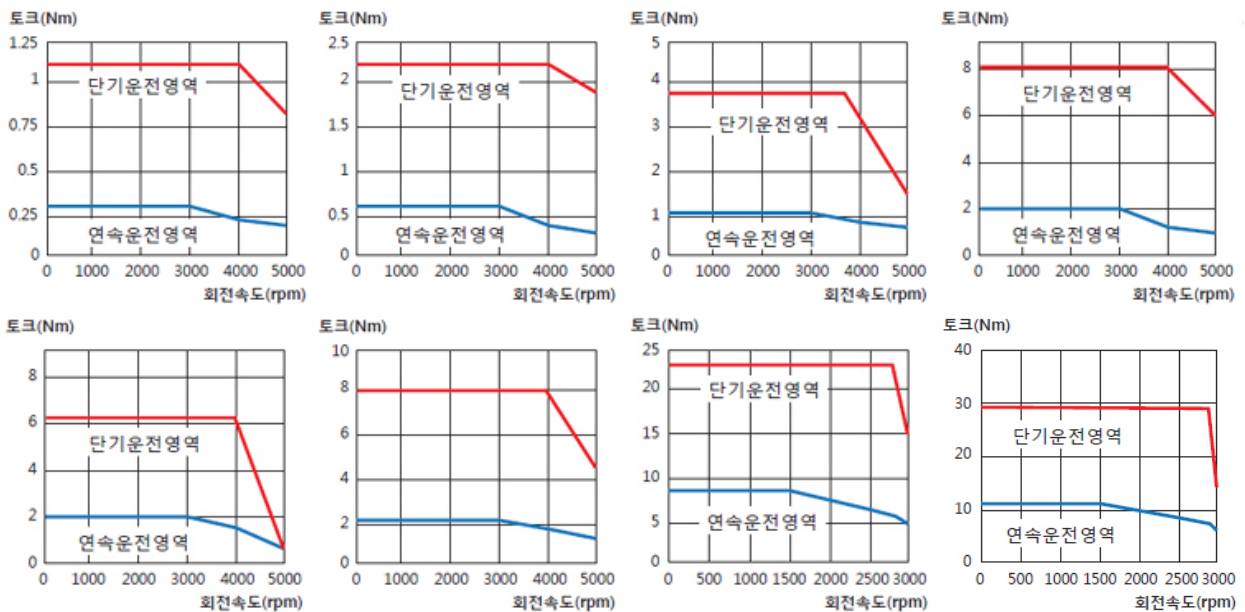
2.1 Servo Motor

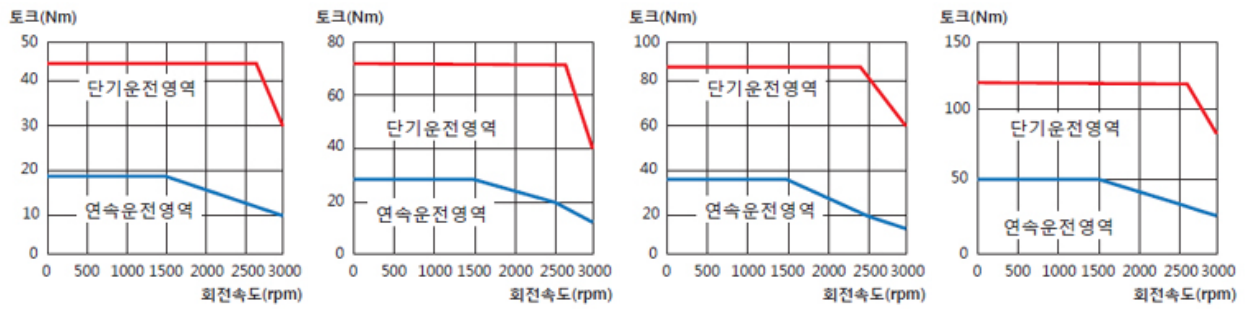
2.1.1 Product Characteristics

Voltage		220V												
Flange Specifications		40	60			80		130			180			
Drive Format	SD-	04A	04A	04A	10A	10A	10A	20A	20A	20A	30A	55A	75A	110A
Motor Type	SMG-SMA-	01A□A	02A□A	04A□A	04A□SA	08A□A	05A□A	13A□A	20A□SA	20A□A	30A□A	44A□A	55A□A	75A□A
Rated Output	kW	0.1	0.2	0.4	0.4	0.75	0.45	1.3	2	1.8	2.9	4.4	5.5	7.5
Rated Torque	N·m	0.312	0.637	1.27	0.91	2.39	2.83	8.3	11.3	11.3	18.4	28.2	34.8	47.8
Peak Torque Torque	N·m	0.1	2.23	4.46	6.69	7.16	8.7	23.0	27.6	28.4	44.7	71.0	87.3	117
Rated Current	Arms	0.84	1.6	2.7	4.2	4.4	3.5	10.4	16	16.5	23.6	32.4	41.8	54.2
Peak Current	Arms	2.9	5.8	9.3	14.9	13.4	10.5	27	42	41	54	83	108	129
Rated Speed Speed	min ⁻¹	3000					1500							
Maximum Rotation Speed	min ⁻¹	5000					3000							
Torque Constant	N·m /Arms	0.352	0.435	0.512	0.505	0.59	0.83	0.85	0.68	0.70	0.82	0.86	0.80	0.90
Rotator Moment of Inertia *1	kg·m ² *10 ⁻⁴	0.0525	0.259	0.442	0.667	0.672	7.22	20.3	26.7	31.4	45.8	67.3	88.7	126
		(0.074)	(0.323)	(0.506)	(0.744)	(0.86)	(9.3)	(22.4)	(34.2)	(40.0)	(54.3)	(75.7)	(97.3)	(133.2)
Rated Power Rate	kW/s	15.2	15.7	36.5	54.7	84.8	11.0	33.5	36.1	41.3	75.1	118	135	183

*1) The values in parentheses indicate values for motors equipped with a holding brake.

※ Speed-Torque Characteristics





2.1.2 External Dimensions

SMA-F60 / F80	
SMG-F130	
SMG-F180	
SMG-F180B	

※ External Dimensions Table

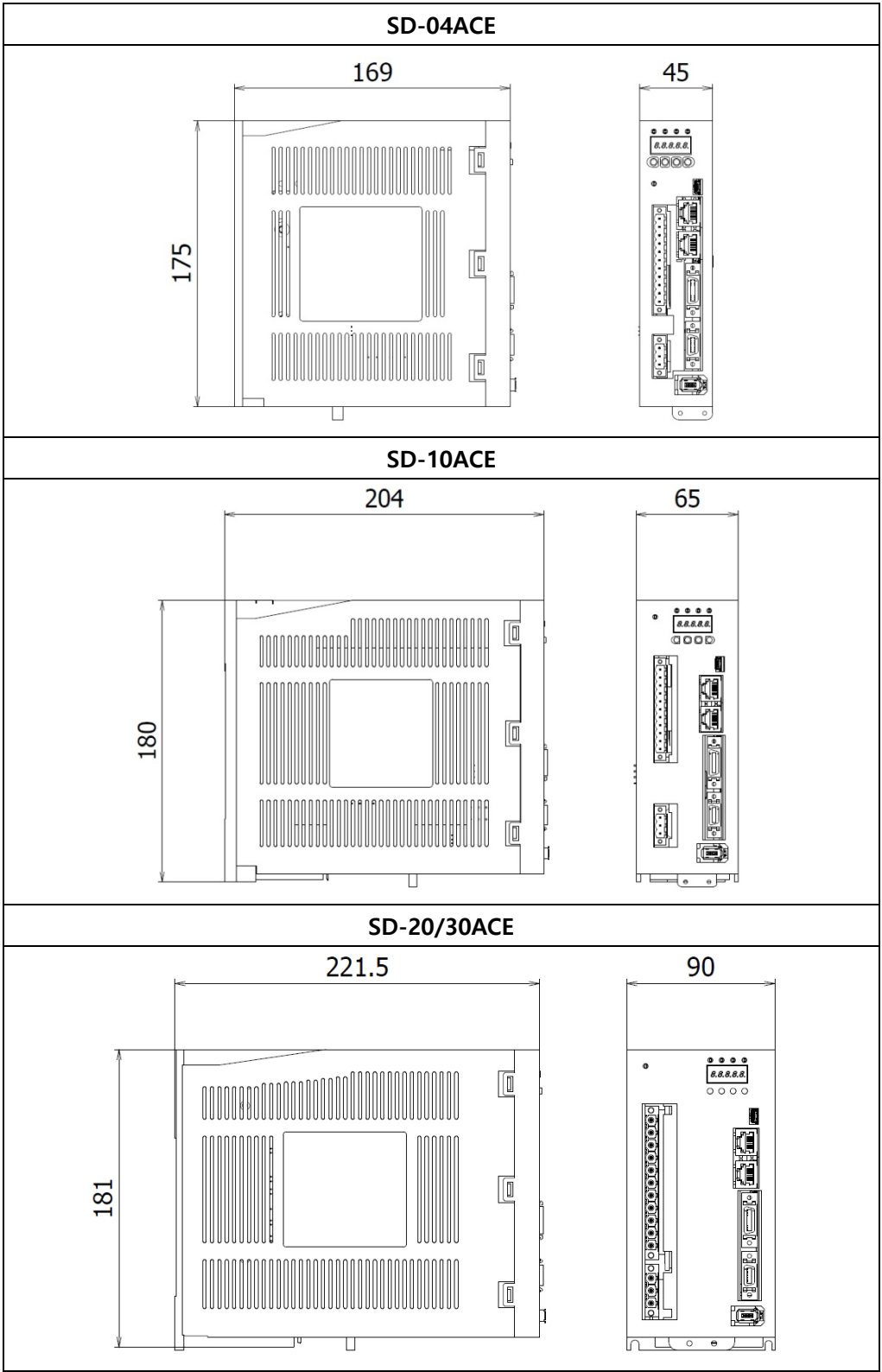
Model	External Dimensions										
	A	B	C	D	E	F	S	R	G	H	I
SMA-01	138	28.5	110	25.5	2.5	65	8	30	46	4.4	40
SMA-01B	170	28.5	146.7	25.5	2.5	96.5	8	30	46	4.4	40
SMA-02	141	28	113	25	3	68	14	50	70	5.3	60
SMA-02B	178	28	150	25	3	105	14	50	70	5.3	60
SMA-04	166	28	138	25	3	93	14	50	70	5.3	60
SMA-04B	203	28	175	25	3	130	14	50	70	5.3	60
SMA- S04	165	40	125	37	3	80	16	70	90	7.2	80
SMA- S04B	186	40	162	37	3	117	16	70	90	7.2	80
SMA-08	197	40	157	37	3	112	16	70	90	7.2	80
SMA-08B	233	40	194	37	3	149	16	70	90	7.2	80
SMG-05	195	56	144	50	6	91	22	110	145	9	130
SMG-05B	234	56	177.5	50	6	137.5	22	110	145	9	130
SMG-13	237	56	180	50	6	133	22	110	145	9	130
SMG-13B	276	56	219.5	50	6	179.5	22	110	145	9	130
SMA-S20	282	66	216	60	6	171	24	110	145	9	130
SMA-S20B	323	66	257	60	6	217	24	110	145	9	130
SMG-20	224	78	165	75	3	121	35	114.3	200	13.5	180
SMG-20B	293	78	215	75	3	173	35	114.3	200	13.5	180
SMG-30	270	78	192	75	3	148	35	114.3	200	13.5	180
SMG-30B	319	78	241	75	3	199	35	114.3	200	13.5	180
SMG-44	304	78	225	75	3	178	35	114.3	200	13.5	180
SMG-44B	353	78	275	75	3	233	35	114.3	200	13.5	180
SMG-55	373	112	260	108	3	213	42	114.3	200	13.5	180
SMG-55B	422	112	310	108	3	270	42	114.3	200	13.5	180
SMG-75	445	112	332	108	3	285	42	114.3	200	13.5	180
SMG-75B	493	112	382	108	3	338	42	114.3	200	13.5	180

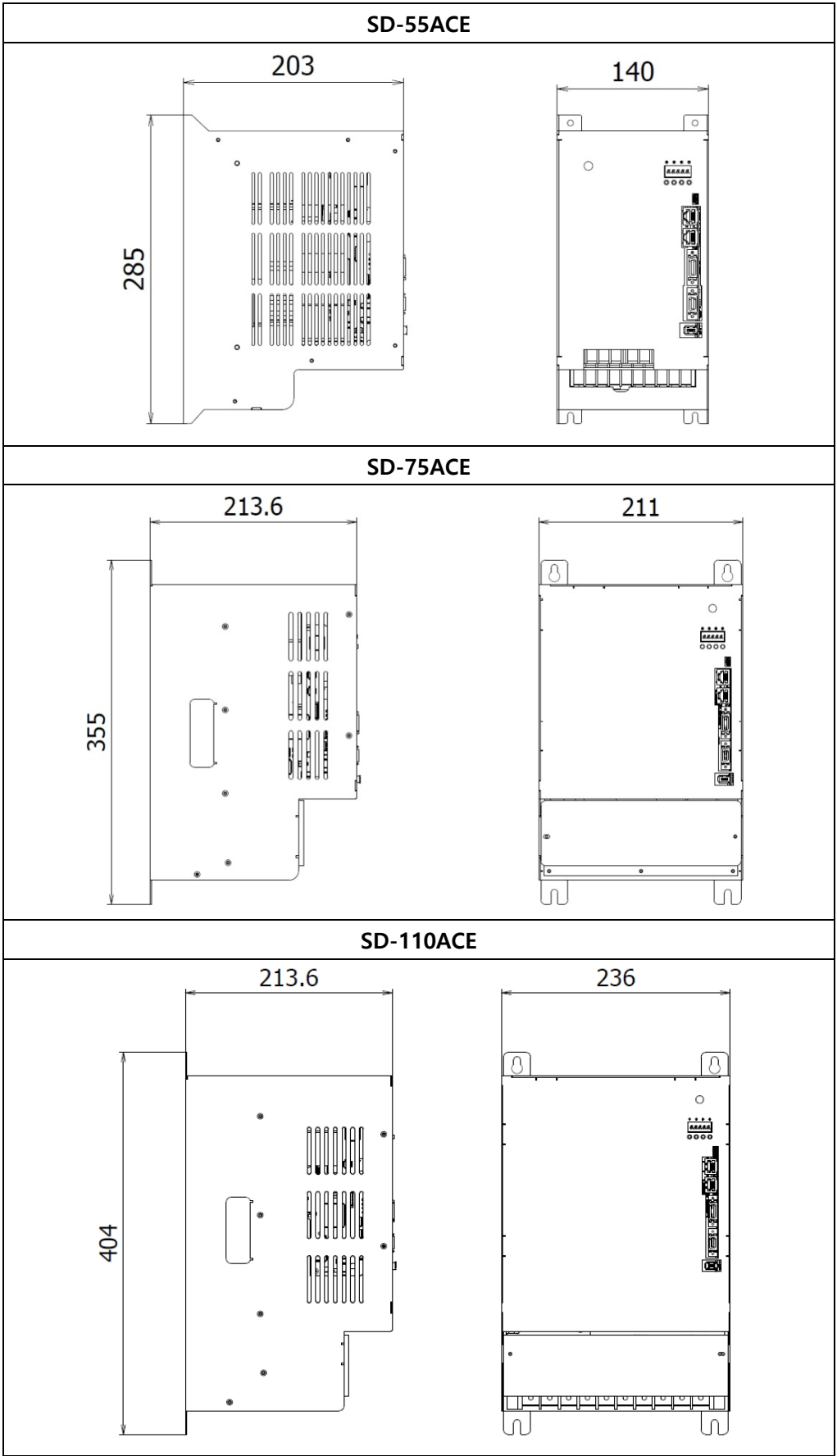
2.2 Servo Drive

2.2.1 Product Characteristics

Servo Pack Type SD-		04A	10A	20A	30A	55A	75A	110A
Maximum Applicable Motor [KW]		0.4	1.0	2.0	3.0	5.5	7.5	11
Rated Output Current [A]		3.0	6.5	13.5	16.5	35.0	40.0	50.0
Maximum Output Current [A]		9.0	19.5	40.5	49.5	96.0	104.0	125.0
Recovery Resistor		Internal / External				External		
Main Circuit [3-phase: AC200V]		AC200 ~ 230V +10~-15% 50/60Hz						
Control Circuit [Single-phase: AC200V]		AC200 ~ 230V +10~-15% 50/60Hz						
Main Power Circuit	Input Voltage Frequency	3-phase 200V ~ 230V, 50/60Hz						
	Allowable Voltage Fluctuation Rate	Within +10% to -15%						
Control Circuit Power supply	Input Voltage	Single-phase 200V to 230V, 50/60Hz						
	Allowable Voltage Fluctuation Rate	Within ±15%						
Detector	Encoder Type	Incremental 2000~6000[ppr] 15-wire, 17-bit serial encoder, 19-bit serial encoder						
	Output Signal Type	Differential Line Driver Output.						
	Detector Accuracy	131072 per revolution (based on serial encoder)						
	Detector Power Supply	DC 5V, Max 0.3[A]						
Drive Method	Drive Method	Sine wave PWM control (using IGBT)						
	GATE Circuit	1ED020I12-F2 (Max 2.0A)						
	Protection Functions	UVLO (Under Voltage Lockout), Ready, Short Circuit Clamping, Active Miller Clamp, Desaturation Protection, Fault Signal						
Speed Control Specifications	Frequency Response Characteristics	Up to 1 kHz or higher						
	Speed Command Input	DC 0V to ±10V (Maximum Speed Parameter Adjustable), EtherCAT Speed Command						
	Speed Variation Rate	±0.01[%], Load Variation 0~100%						
	Acceleration/deceleration time	Linear or S-curve acceleration/deceleration						
Position Control Specifications	Position input frequency	1Mpps, Line Drive						
	Position Input Type (Pulse)	Direction + Pulse, Clockwise Pulse + Counterclockwise Pulse						
	Position Input Format	Line Driver Method (RS422)						
Torque Control Specifications	Torque Command Input	DC 0V to ±10V, maximum speed parameter adjustable)						
	Repeatability	Within ±1%						
	Torque Limit Speed Command	DC0V~±10V, maximum speed parameter adjustable)						
Built-in Function	Protection Functions	Overcurrent, Regenerative Overvoltage, Overload, Encoder Abnormality, Main Power Abnormality, Overheating, excessive regeneration, overspeed, IGBT FAULT, undervoltage, etc.						
	Regenerative Resistor	Built-in standard, external (separate)						
	USB Communication	PC USB communication supported (Parameter and variable monitoring)						
	Temperature Detection	Motor temperature, IGBT temperature, PCB temperature detection						

2.2.2 External Dimensions





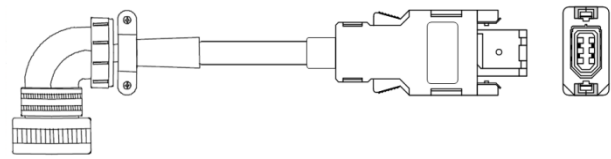
2.3 Peripheral Devices

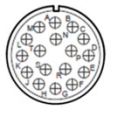
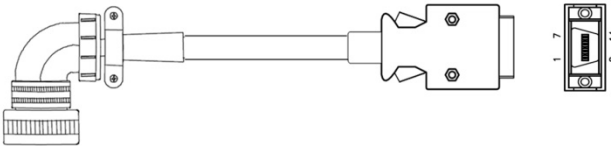
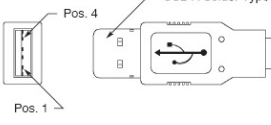
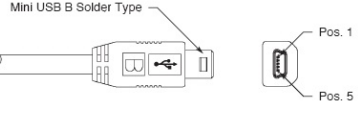
2.3.1 Braking Resistor

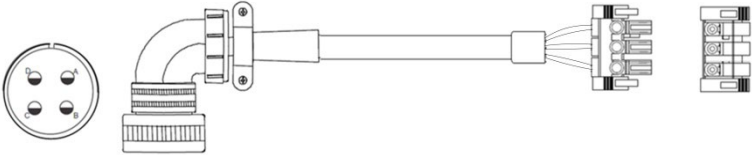
Category	Product Name	Model	Applicable Drive	Specifications
Resistance	IO Cable	APC-300R23	SD-04AC□ SD-10AC□	
Resistor	IO Cable	APC-600R30	SD-20AC□ 2P SD-30AC□ 3P SD-55AC□ 3P SD-75AC□ 3P SD-110AC□ 4P	

2.3.2 Cables and Connectors

Category	Product Name	Model	Specifications
Signal	IO Cable		Upper Controller Drive Connection (CN1) 핀 번호 표시 Drive Connection - Specification (case): 10320-52A0-008 (3M) - Specification (Plug): 10120-3000PE (3M)

Signal	Encoder Cable (Serial)		Motor Connection  Drive Connection (CN2) 
			Motor connector specifications: MS310B 20-29s Drive Connector Specifications: 59233-6003 (Molex)

Category	Product Name	Model	Specifications
Signal	Encoder Cable (Serial)		Motor Connection Port  Drive Connection Port (CN3) 
			Motor Connection - Specifications: MS310B 20-29s Drive Connector - Specification (case): 10314-52A0-008 (3M) - Specification (Plug): 10114-3000VE (3M)
Signal	Assembly Cable		PC Connector  Drive Connector (CN6) 
			PC Connector, Drive Connector - Specification: 88732-8900 (Molex)

Power		Motor Connector Connector  Motor Connector - Specifications: MS310B 20-4S Drive Connection - Specifications: BLZ 7.62HP/03/180LR SN BK BX SO (Binder Müller) Drive
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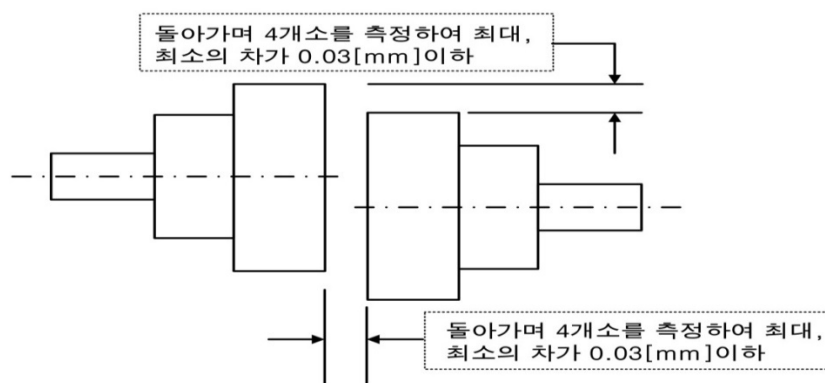
3.1 Servo Motor

3.1.1 Operating Environment Conditions

Ambient Temperature	0~40°C (No freezing)
Ambient Humidity	80% RH or less (no condensation)
External Vibration	X, Y: 19.6[m/s ²] or less

3.1.2 Preventing Shaft Impact

It is important to align the motor shaft center with the shaft center of the mating machine. Misalignment causes vibration and may damage the bearings. When installing the coupling, use a rubber mallet or similar tool to cushion impacts and prevent excessive force from acting on the shaft and bearings.



3.1.3 Mounting Surface Tolerances

The table below indicates the accuracy of the output shaft and mounting surfaces for AC servo motors.

Item	Tolerance (T.I.R)	Reference Drawing
Perpendicularity between flange mounting surface and output shaft ①	0.04 mm	
Flange fitting outer diameter eccentricity ②	0.04 mm	

Output shaft end runout ③	0.02 mm	
---------------------------	---------	--

3.1.4 Installation Orientation

- Installation is possible horizontally, with the shaft positioned at the top or bottom.
- Ensure cables from the motor are oriented downward.
- When installing vertically, install a cable trap to prevent oil and water from flowing into the motor.

3.1.5 Cable Disconnection

- Take care not to stress or damage the cable.
- When using the motor for moving applications, always use a flexible cable.

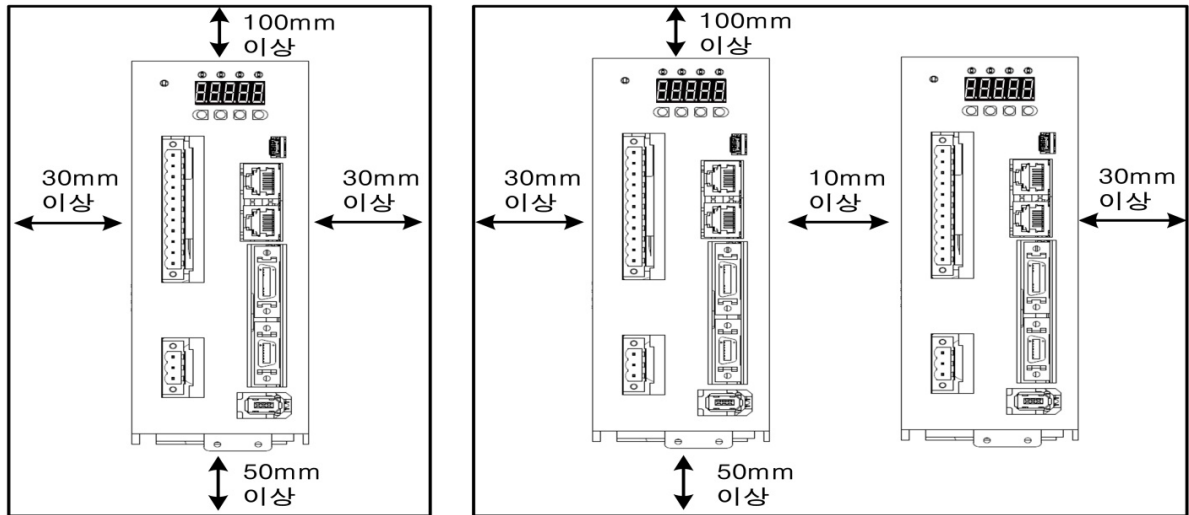
3.2 Servo Drive

3.2.1 Operating Environment Conditions

Ambient Temperature	0~50°C (No freezing)
Ambient Humidity	90% RH or less (no condensation)
External Vibration	X, Y: 4.9[m/s ²] or less

3.2.2 Installation Orientation and Spacing

- 1) Install the drive so that the front panel faces forward.
- 2) When installing the drive in an enclosed control panel, maintain a minimum spacing of 10mm between drives horizontally and 40mm vertically. Especially when installing multiple units in parallel, a vertical clearance of approximately 100mm is required. Ensure fans or similar components are not obstructed to prevent heat buildup.



3.3 Handling

3.3.1 Handling Precautions

Improper handling may cause unexpected accidents or damage. Please use correctly by referring to the following points.

1) Handling

- Do not apply impact to the encoder, which is the motor detector. Striking the shaft with a hammer or dropping the motor may cause damage.
- Do not directly connect the motor to commercial power (AC380V). Excessive current flow will degrade the motor's magnetic performance. Always drive it in combination with the specified servo drive.

2) Wiring

- Connect the ground terminals of the drive and motor to the drive side and ground them using the shortest possible path. Use a Type 3 ground (100Ω or less) to prevent electric shock and malfunction.
- The U, V, W, and FG terminals of the drive and motor must be matched exactly. Unlike commonly used universal motors, the rotation direction cannot be changed by swapping two wires.
- Connecting commercial power to the drive's U, V, W, FG terminals may cause damage. Apply 400V-rated power to the R, S, T terminals. For power sources other than 400V-rated, a transformer must be installed.
- A standard regenerative resistor must be connected to the drive's B1+ and B3 terminals.

Chapter 4

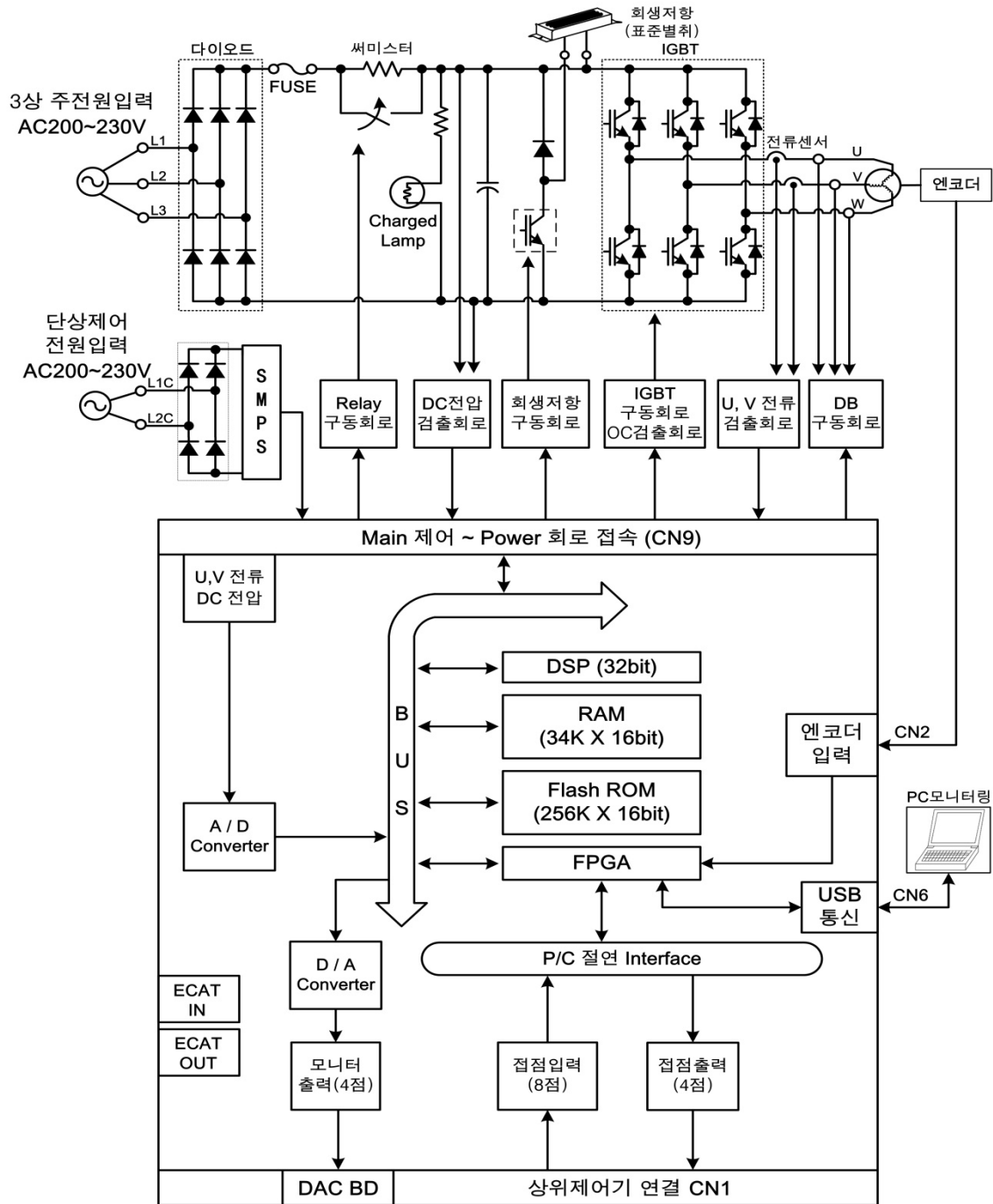
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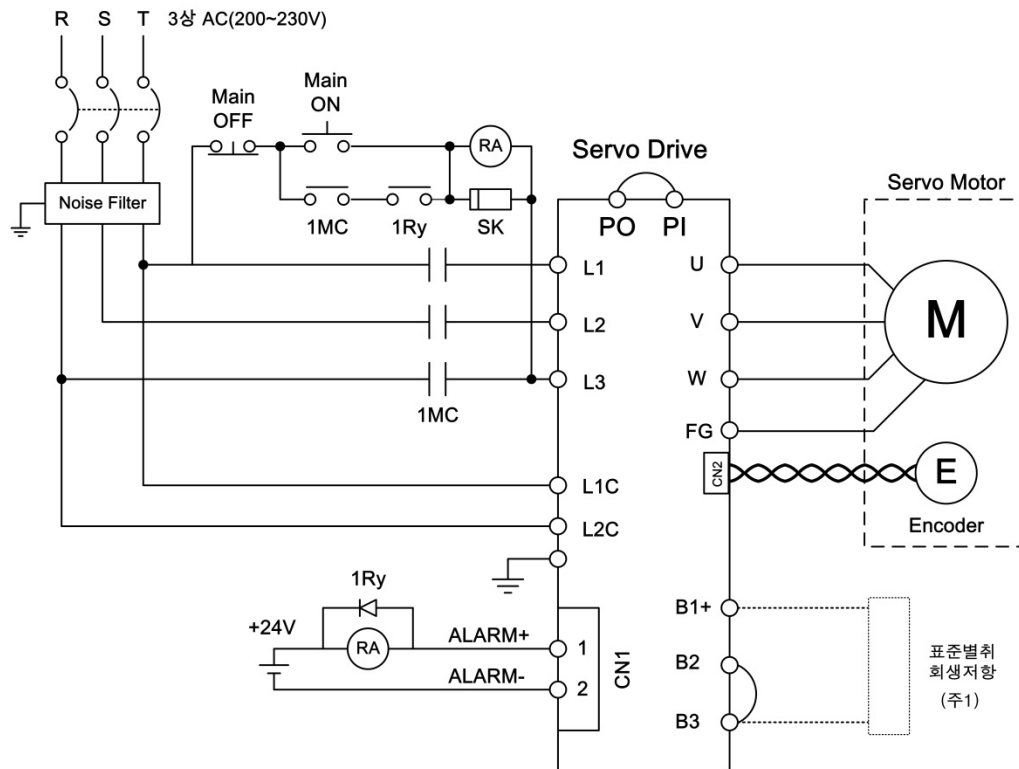
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4.1 Internal Block Diagram



4.2 Power Supply Wiring



(Note 1): If frequent acceleration/deceleration results in high regenerative capacity, use a regenerative resistor with the same resistance value but higher capacity than the standard resistor provided.

4.3 Power Circuit Electrical Component Specifications

Model	SD-04AC□	SD-10AC□	SD-20AC□	SD-30AC□	SD-55AC□	SD-75AC□	SD-110AC□
MCCB	ABS33bM (10A)	ABS33bM (10A)	ABS33bM (20A)	ABS33bM (30A)	ABS33bM (50A)	ABS33bM (60A)	ABS33bM (80A)
Noise Filter	TB6- B030NBDC	TB6- B030NBDC	TB6- B030NBDC	TB6- B030NBDC	TB6- B060NBDC	TB6- B080NBDC	TB6- B080NBDC
MC	GMC-40 (35A)	GMC-40 (35A)	GMC-40 (35A)	GMC-40 (50A)	GMC-40 (60A)	GMC-40 (100A)	GMC-40 (100A)
Wire	AWG12 (3.5SQ)	AWG12 (3.5SQ)	AWG12 (3.5SQ)	AWG12 (5.5SQ)	AWG12 (5.5SQ)	AWG12 (8.0SQ)	AWG12 (8.0SQ)
Regenerative Resistor (Included)	50W 50Ω	140W 40Ω	300W 23Ω	300W 2P 11.5Ω	300W 2P 11.5Ω	300W 2P 11.5Ω	Option

4.4 Noise Countermeasures

Wiring and grounding for the servo drive must be configured to minimize the impact of switching noise generated by the IGBT device. Care must also be taken to prevent interference from external noise sources as much as possible.

Grounding

Since the servo drive supplies power to the motor based on the switching of the IGBT device, a current $C_f dv/dt$ flows through the motor's stray capacitance (C_f) from the power section. To prevent the effects of this switching noise, the motor frame terminal must be connected to the FG terminal inside the servo drive terminal block. Furthermore, the servo drive FG terminal must be directly grounded to the reference ground plane.

Noise Filter

Use a noise filter to prevent noise from the power line. Install it considering the following points:

Keep input and output wiring separate; do not bundle wires or place them in the same conduit.

Do not place the ground wire in the same duct or bundle it with the filter output wire or other signal wires.

The ground wire must be wired separately to the ground plate.

When a filter is installed within a unit, ground the filter and the unit's internal equipment to the unit's base.

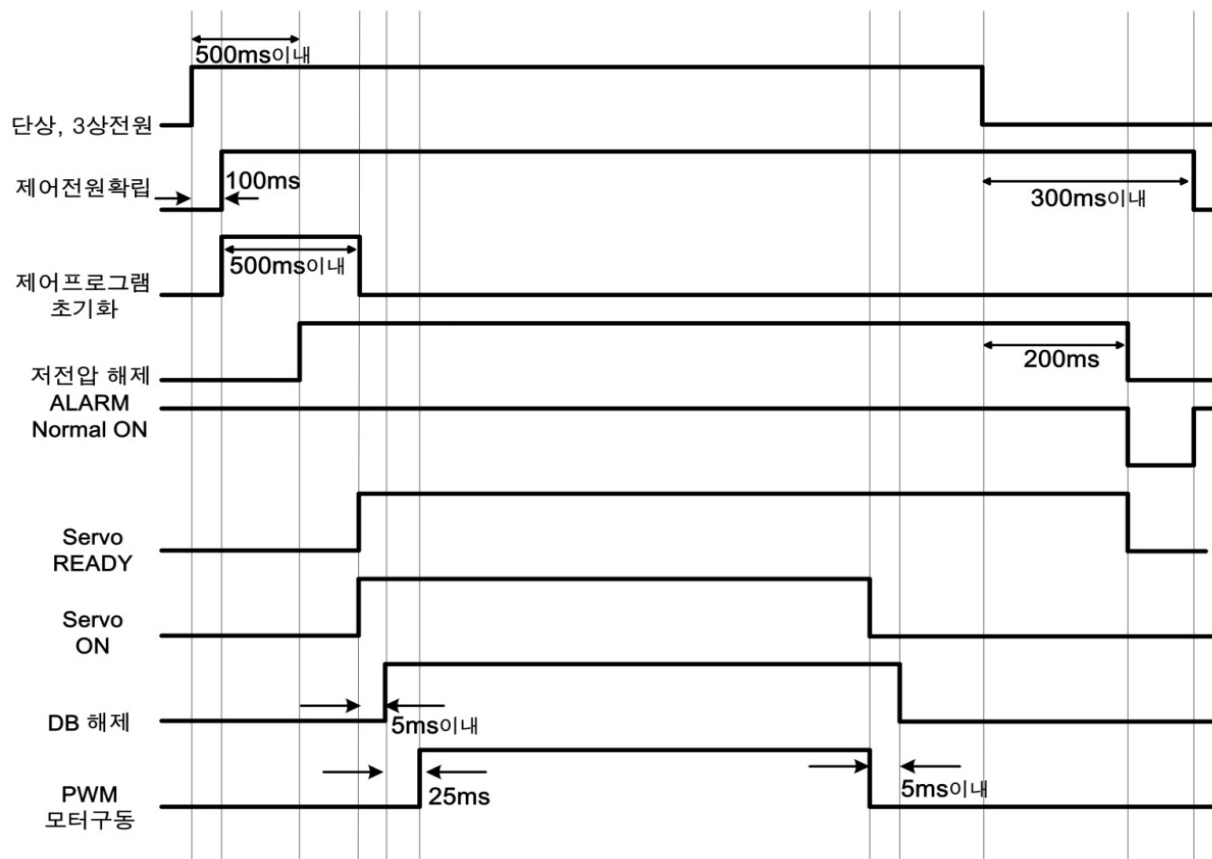
Use noise filters specified for electrical components or equivalent filters.

4.5 Timing Diagram

4.5.1 Power-On Timing Diagram

When AC single-phase 220V power is connected to the L1C and L2C terminals of the servo drive, power is supplied to the control circuit. When AC three-phase 220V power is connected to L1, L2, and L3, power is supplied to the main circuit.

After a maximum of 500[ms], which is the time required for initialization inside the drive unit, the servo becomes ready (RDY). When the servo drive signal is turned ON, operation begins after 30[ms].

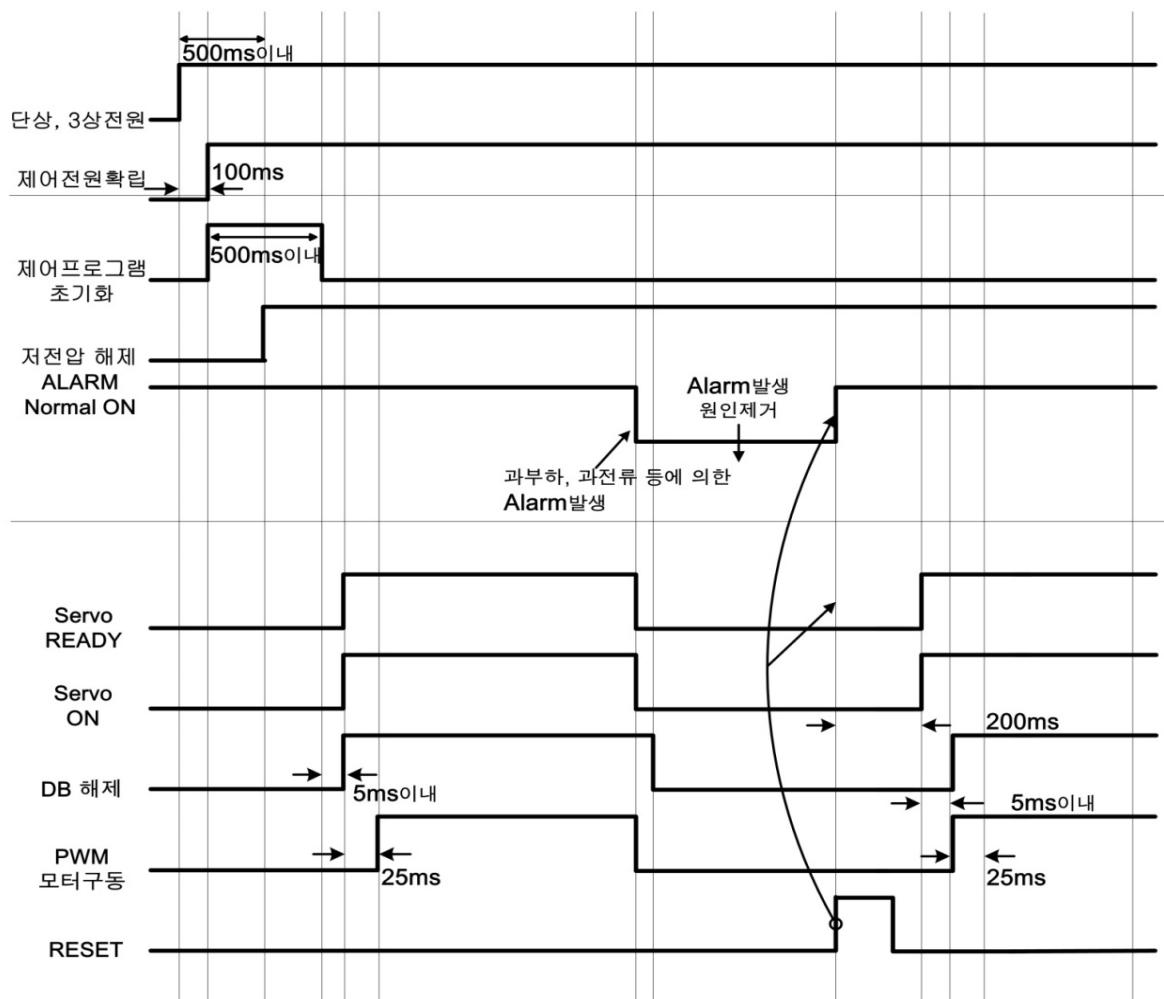


4.5.2 Timing During Alarm Occurrence

When an alarm occurs in the servo drive, PWM is blocked, and the motor stops.

To reset the alarm, eliminate the cause, turn OFF the servo motor drive command (SVON) signal, and then

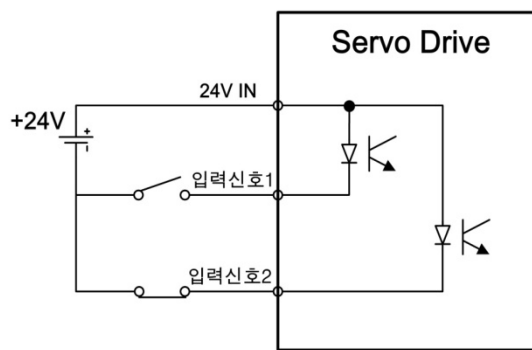
4.6 Control Signal Wiring



4.6 Control Signal Wiring

4.6.1 Contact Input Signal

Contact Input Signal Wiring Example

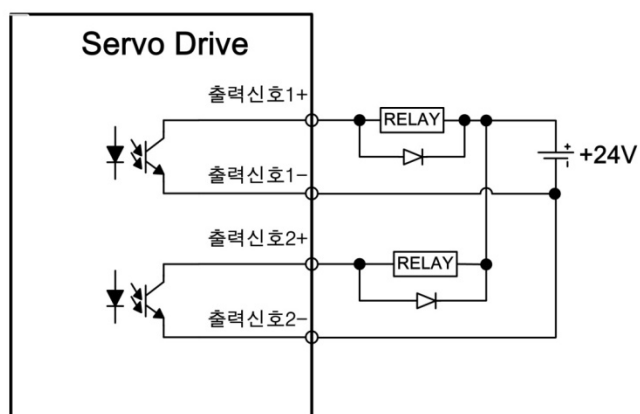


4.6.2 Contact Output Signals

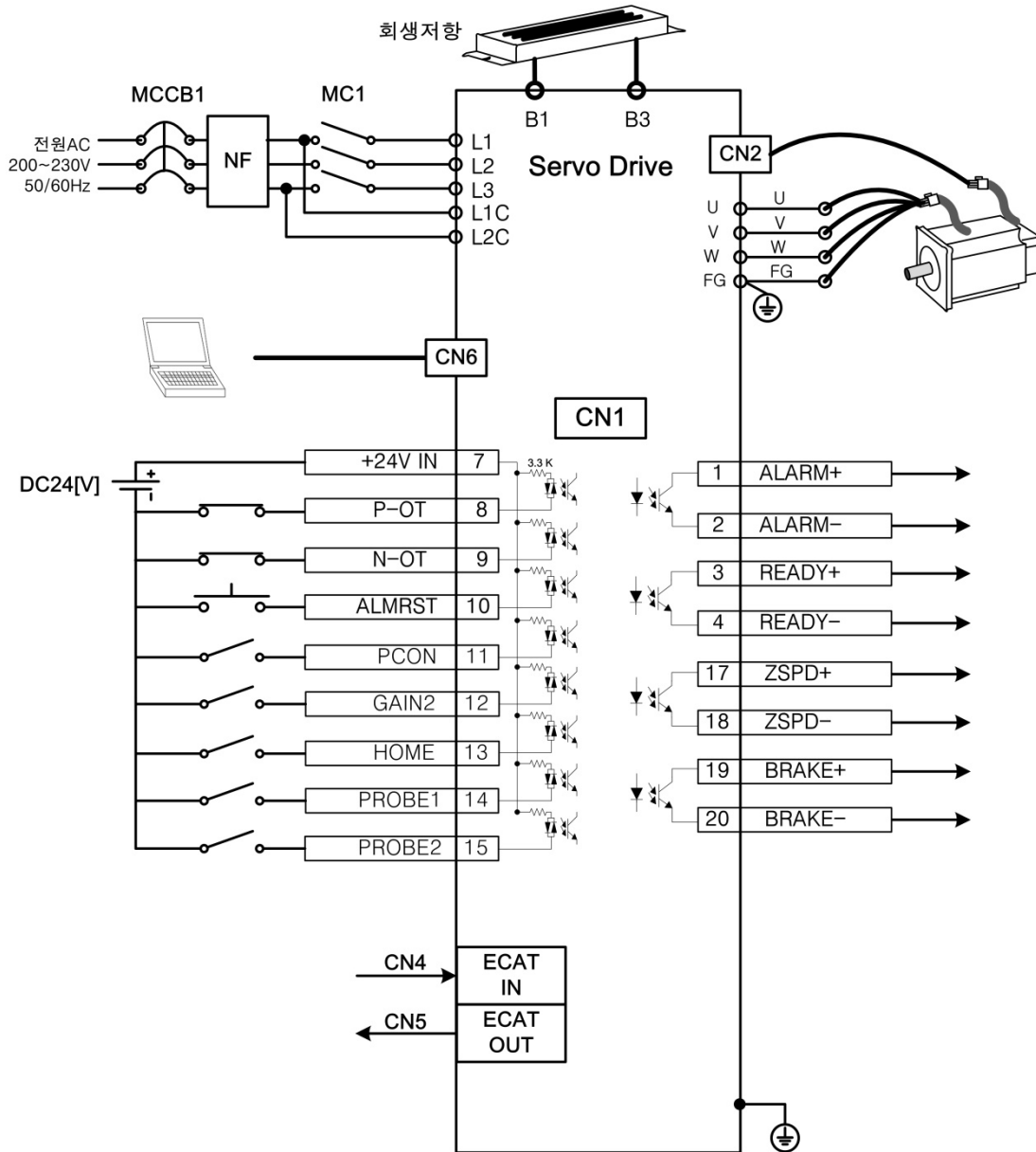
The output contacts use internal transistor switches; therefore, overvoltage or overcurrent may cause damage.

. Use with caution.

- Rated Specifications: DC24[V] $\pm 10\%$, 150[mA]



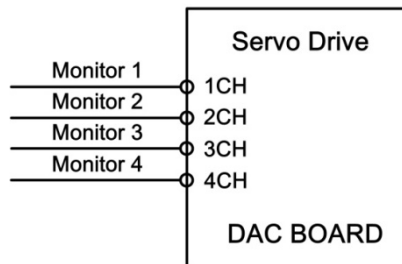
4.7 Input/Output Signal (CN1) Wiring



- 1) NF stands for Noise Filter. It must be used to prevent noise intrusion.
- 2) The regenerative resistor is built-in and mounted on the bottom of the drive heat sink.
- 3) Connect the ground wire to the drive's FG (Frame Ground) terminal.

4.8 Analog Output Signal Wiring (Optional)

Analog output signals can be utilized by connecting a separate DAC board to the drive unit. The DAC board allows monitoring of 4 channels.



- 1) Set GND to the control power supply's 0[V].
- 2) The monitor output signal voltage is +5[V].

4.9 Incremental Encoder Signal Section (CN3) Wiring

CN3 is a connector located at the bottom front of the drive unit.

This connector is used to connect the encoder of the servo motor to the drive unit. The pin arrangement as viewed from the user-side connector is as shown in the figure below.

Encoder signals may vary depending on the type of encoder.

2	/PA	1	PA	9	PV	8	/PU
4	/PB	3	PB	11	PW	10	/PV
6	/PZ	5	PZ	13	VCC	12	/PW
		7	PU			14	GND

[Based on the solder side of the user connector]

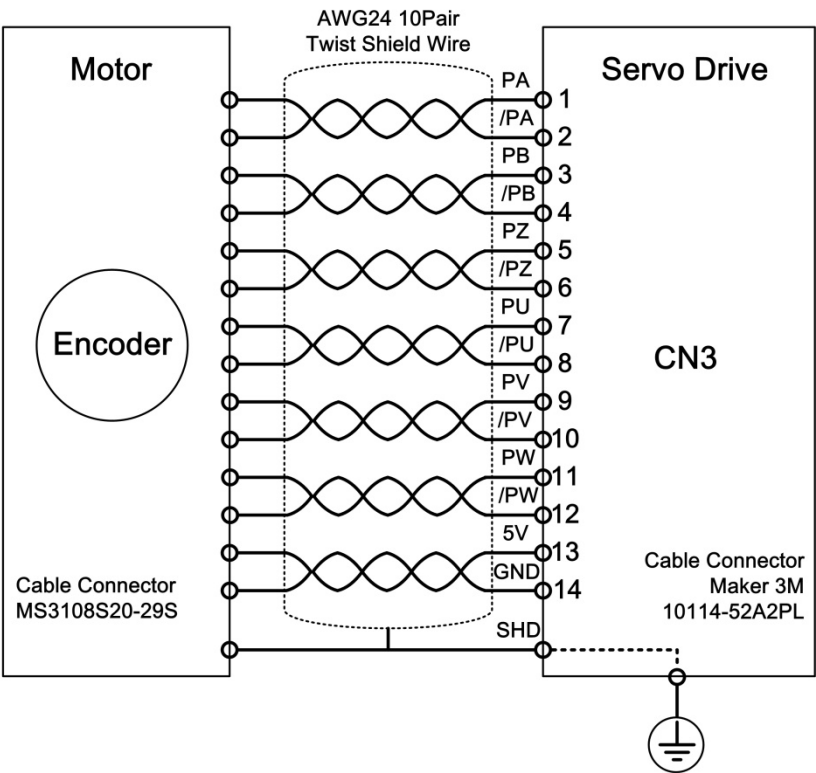
■ Manufacturer: 3M, Part Number: 10114-52A2PL

The wiring for the incremental encoder on CN3 is as shown in the table below.

CN3 PIN No.	Signal Name	MOTOR Side Encoder Connector PIN No.
1	PA	
2	/PA	
3	PB	
4	/PB	
5	PZ	
6	/PZ	
7	PU	
8	/PU	
9	PV	
10	/PV	
11	PW	
12	/PW	
13	Vcc (DC 5V)	
14	GND	
Shield	FG	

■ Applicable Cable Specifications: AWG24 × 10Pair TWIST SHIELD CABLE (Maximum Length 20m)

Subdrive CN3 Wiring Example for Incremental Encoder Application



4.10 17-bit Absolute/Incremental Encoder Signal Section (CN2) Wiring

CN2 is the connector located at the top of the lower section of the drive unit. This connector is used to connect the drive unit to the 17-bit serial encoder of the servo motor. The pin arrangement as viewed from the user-side connector is shown in the figure below. Encoder signals may vary depending on the encoder type.

2	GND	1	Vcc
4	BAT-	3	BAT+
6	/SD	5	SD

[Based on the solder side of the user connector]

■ For absolute 17-bit encoders, connect an external battery to the encoder's BAT+ and BAT- terminals.

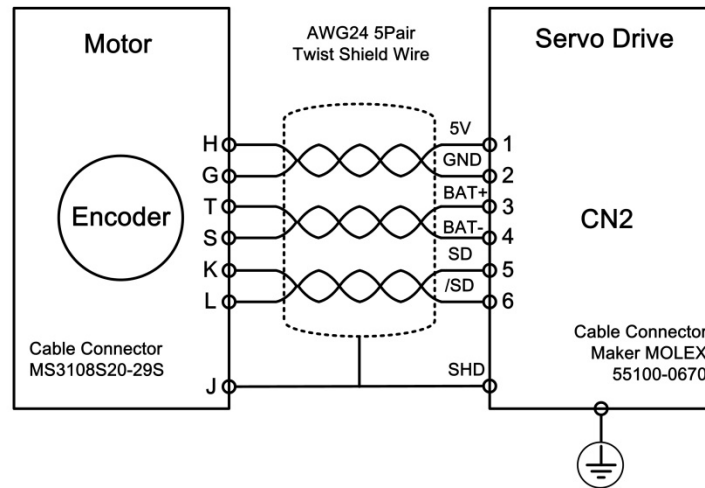
■ Manufacturer: MOLEX, Part Number: 55100-0670

The wiring for the 17-bit absolute encoder between CN2 and the SM□ series AC servo motor is as shown in the table below.

CN2 PIN No.	Signal Name	MOTOR-side Encoder Connector PIN No.
1	Vcc (DC 5V)	H
2	GND	G
3	BAT+	T
4	BAT-	S
5	SD	K
6	/SD	L
Shield	FG	J

■ Applicable Cable Specifications: AWG24 × 5 Pair TWIST SHIELD CABLE (Maximum Length 20m)

SM□ Series AC Servo Motor and Servo Drive CN2 Wiring Example for 17-bit Absolute/Incremental Encoder Application



■ For 17-bit incremental encoders, the encoder's BAT+ and BAT- terminals do not need to be connected.

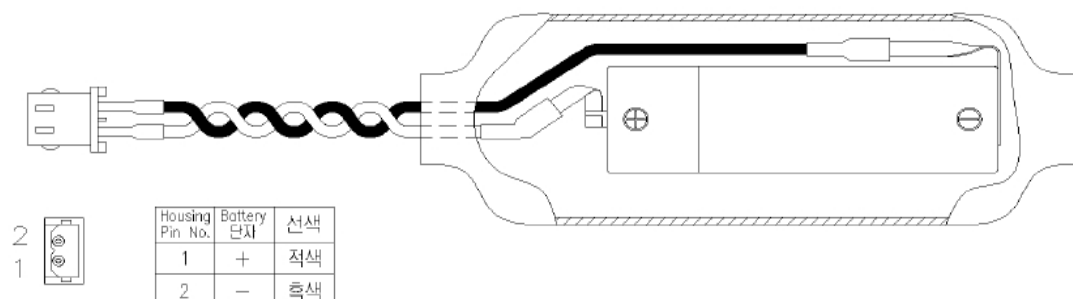
■ External Battery

Product Name: Lithium Battery (ER6V/2.6V,
2Ah)/SB-AA11

Manufacturer: TOSHIBA

Product Name: Lithium Battery (SB-
D002)/3.6V/1.9Ah

Manufacturer: Tekcell



4.11 19-bit Absolute/Incremental Encoder Signal Unit (CN3) Wiring

CN3 is a connector located at the bottom of the drive unit. This connector is used to connect the drive unit to the servo motor's 19-bit serial encoder. Encoder signals may vary depending on the encoder type.

■ For an absolute 19-bit encoder, connect an external battery to the encoder's BAT+ and BAT- terminals.

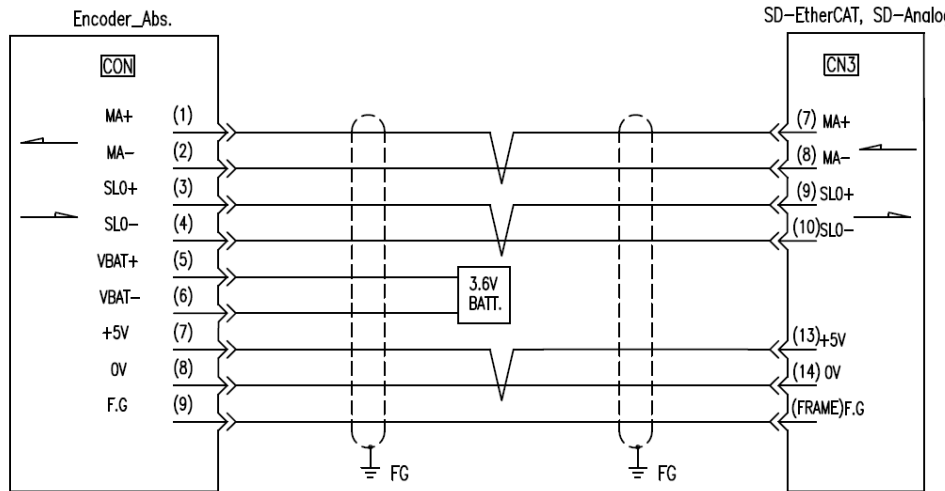
■ Manufacturer: MOLEX, Part Number: 55100-0670

The wiring for the 19-bit absolute encoder between CN3 and the SM□ series AC servo motor is as shown in the table below.

CN3 PIN No.	Signal Name	MOTOR-side Encoder Connector PIN No.
7	MA+	1
8	MA-	2
9	SL0+	3
10	SL0	4
	BAT+	5
	BAT-	6
13	Vcc (DC 5V)	7
14	GND	8
Shield	FG	9

■ Applicable Cable Specifications: AWG24 × 5Pair TWIST SHIELD CABLE (Maximum Length 20m)

SM□ Series AC Servo Motor and Servo Drive CN2 Wiring Example for 19bit Absolute/Incremental Encoder Application



■ For the Incremental 19it encoder, the encoder's BAT+ and BAT- terminals do not need to be connected.

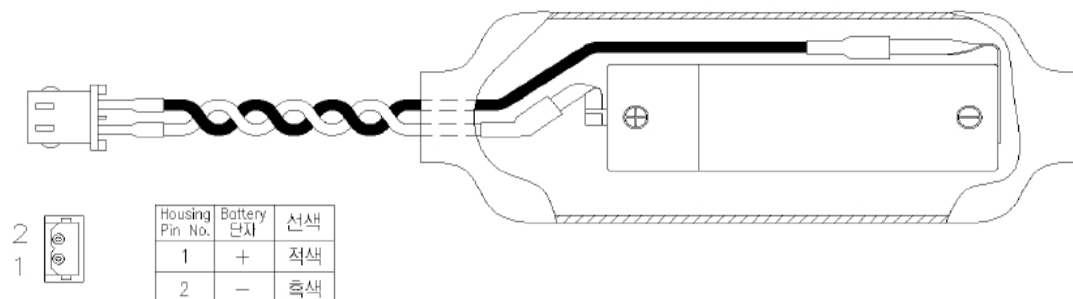
■ External Battery

Product Name: Lithium Battery (ER6V/2.6V,
2Ah)/SB-AA11

Manufacturer: TOSHIBA

Product Name: Lithium Battery (SB-
D002)/3.6V/1.9Ah

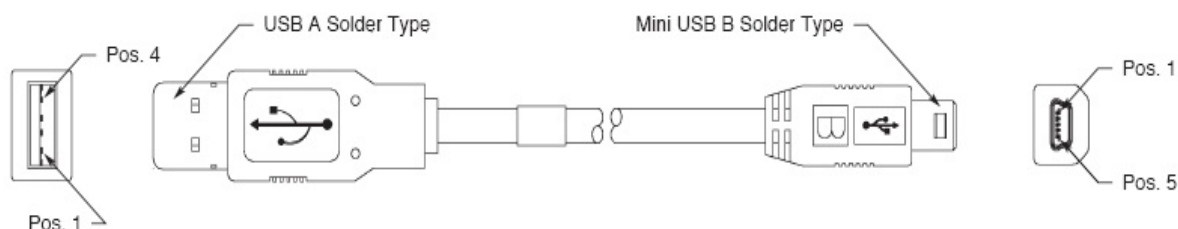
Manufacturer: Tekcell



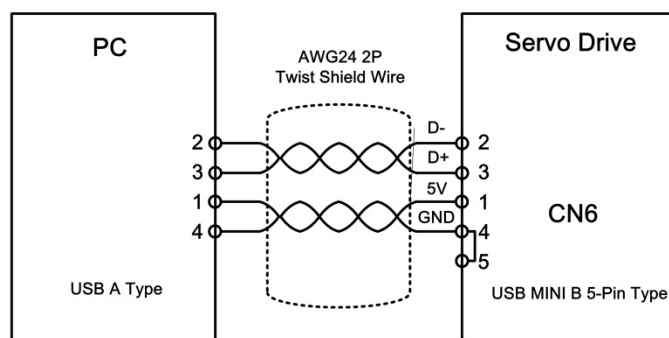
4.12 USB Monitoring Signal Unit (CN6) Wiring (Optional)

CN1 is the connector located at the center front of the drive unit. This connector is used to connect the drive unit to a PC (Personal Computer). Using a dedicated program on the PC, you can change the parameters of the motor and drive via USB serial communication and perform real-time monitoring tasks. The CN6 connection terminal on the drive unit is a USB Mini B 5-Pin Male Type.

The figure below shows the standard drawing and pin map for the USB A Type - Mini USB B Type Assembly cable connecting the drive unit to the PC.



[USB A Type - Mini USB B Type Assembly Drawing]



4.13 EtherCAT Cable Wiring Method (CN4/CN5)

CN4/CN5 are EtherCAT ports located at the top center of the servo drive. They are used for connecting masters and slaves for EtherCAT communication.

4.13.1 EtherCAT Connector Wiring

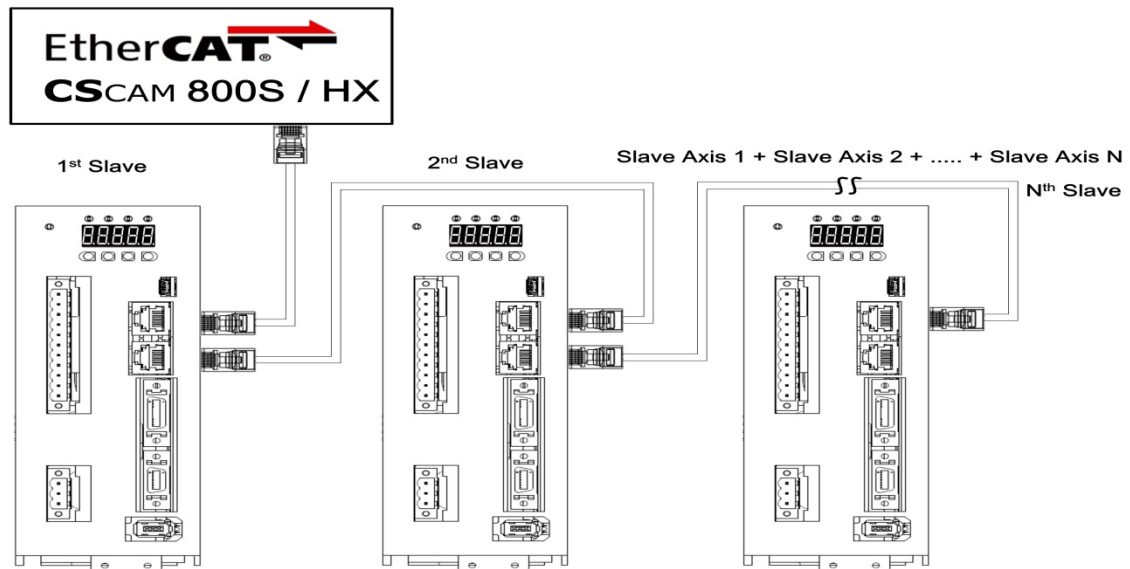
The table below shows the pin map for the connectors connected to the EtherCAT port.

IN / OUT PORT		Function
CN4		EtherCAT Input
CN5		EtherCAT Output
Pin No.	Signal	Line Color
1	TX+	White/Orange
2	TX-	Orange
3	RX+	White/Green
4	-	-
5	-	-
6	RX-	Green
7	-	-
8	-	-
Plate		Shield

- Ethercat communication uses only signal pins 1, 2, 3, and 6.
- Recommended connector: STP CAT.5E RJ-45 8P8C
- Recommended cable: CAT.5 STP

4.13.2 EtherCAT Connection Method

The following diagram shows the connection between the master and slave, and between slaves for EtherCAT communication.



Chapter 5

Maintenance

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5.1 Maintenance and Inspection

This chapter explains basic maintenance and inspection methods for servo motors and drives, as well as fault diagnosis and countermeasures.

5.1.1 Precautions

- 1) When measuring motor voltage: The voltage output from the servo drive to the motor is PWM-controlled, resulting in a pulsed waveform. Measurement values can vary significantly depending on the type of instrument. For accurate measurement, always use a rectified voltmeter.
- 2) When measuring motor current: The pulse waveform may be smoothed into a sine wave due to motor reactance. Therefore, directly connect and use an armature-type ammeter.
- 3) When measuring power: Measure using the three-phase power meter method with a current-voltage meter.
- 4) Other instruments: When using an oscilloscope or multi-digital meter, do not connect them to ground. Keep the instrument input current below 1 [mA].

5.1.2 Inspection Items

When performing inspections, residual voltage may remain in the internal smoothing capacitor, posing a risk of electric shock.

Therefore, always wait until the Charged Lamp has turned off before proceeding with any inspection.

1) Servo Motor Inspection

Inspection Items	Inspection Interval	Inspection and Maintenance	Remarks
Check for Vibration and Noise	Monthly	Check by touch and sound	No significant increase compared to normal than usual.
Visual Inspection	Depending on the level of contamination and damage	Clean with cloth or air	
Insulation resistance measurement	At least once a year	Disconnect from the drive Measure insulation resistance with a meter. If 10[MΩ] or higher, normal	If below 10[MΩ], request technical Request technical support
Oil seal replacement	At least once every 5,000 hours	Remove from machine and replace	For motors with oil seals

Comprehensive Inspection	At least once every 20,000 hours or once every 5 years	Request technical support	Do not disassemble and clean the servo motor yourself. disassemble and clean the servo motor.
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2) Servo Drive Inspection

The table below summarizes the inspection methods for the servo drive. Please inspect it at least once a year.

Inspection Item	Inspection Frequency	Inspection Procedure	Action if Abnormal
Cleaning the main unit and circuit board	At least once a year	Ensure no dust or oil residue remains Ensure no dust or grease remains	Clean with compressed air or a cloth
Check screw tightness	At least once a year	Terminal blocks, connectors, and fastening screws are not loose	Tighten with a screwdriver or wrench
Check for abnormalities in the main unit or circuit board No component abnormalities	At least once a year	No discoloration, damage, or broken wires due to heat No issues	Technical Support Request

5.1.3 Component Replacement Cycle

The following components may experience performance degradation, damage, or failure due to mechanical friction or aging. Therefore, regular inspection and scheduled replacement are necessary for preventive maintenance.

1) Smoothing capacitors

Characteristics degrade due to effects like ripple current. While capacitor lifespan may vary based on ambient temperature and usage conditions, continuous operation typically has a 10-year standard. Capacitor degradation can accelerate rapidly over a certain period, so perform inspections at least once a year.

- Visual Inspection Criteria

Case condition: Side or bottom case expansion

Condition of cover plate: Noticeable expansion, severe cracks, breakage

Condition of the explosion-proof surface: Significant expansion of the edges, signs of operation

Regularly inspect other external aspects, such as exterior cracks, breaks, discoloration, and electrolyte leakage. The capacitor's lifespan is determined when its rated capacity drops below 85%.

2) Relay Types

Contact failure may occur due to contact wear caused by switching currents. Since the impact of power capacity is significant, the cumulative switching count (switching life) of 100,000 cycles is used as the service life standard.

3) Motor Bearings

Please replace them after 20,000 to 30,000 hours of operation at rated speed and rated load.

Motor bearings are greatly affected by operating conditions. If abnormal noise or vibration is detected during inspection, please replace them.

[Standard Replacement Intervals for Components]

Part Name	Standard Replacement Interval	Replacement Method
Smoothing Capacitor	7–8 years	Replace (Decide after inspection)
Relays		Decide after inspection
Fuse	10 years	Replacement
Electrolytic capacitor	5 years	Replacement with new board (determined after inspection)
Cooling fan	4 to 5 years	Replacement
Motor Bearing		Decided after inspection
Motor oil seal	5,000 hours	Replace

5.2 or higher Diagnosis and countermeasures

If an abnormality occurs during operation, it will be displayed as AL-□□ or Err-□ in the status information display window.

In this case, please take appropriate action using the methods below. If the issue persists, please contact our service department.

5.2.1 Servo Motor

[Cause of Abnormality, Inspection Guidelines, and Action]

Symptom	Cause	Inspection Procedure	Action
Motor does not move.	CCWLIM, CWLIM input is OFF.	Refer to system configuration	Turn on the CCWLIM and CWLIM inputs. ON.
	Menu misconfiguration	Check the menu settings for motor, encoder, encoder type, control mode, etc. Check menu settings	Reset menu
	Motor Defect	Measure motor lead terminals with a tester (Resistance between each phase: several ohms)	Replace the motor.
	Loosening of fastening screws	Inspect fastening points	Tighten any loose parts.
	External wiring, cables Open circuit	Inspect motor and encoder wiring.	Re-wire. Replace the cable.
	Encoder malfunction	Check the output waveform.	Replace the encoder. (Request technical support)
	Poor connection	Check the motor lead terminal connections.	Repair the faulty part.
	Input voltage is low	Check the drive input voltage.	Change the power source.
	Overload condition.	Check the machine condition.	Remove foreign objects from rotating parts and Supply lubricant.
The motor overheats.	Ambient temperature is high.	Check the ambient temperature around the motor installation area. (Below 40°C)	Modify the heat dissipation structure. Install a cooling fan.

	Contamination on the motor surface	Check for foreign matter adhering to the motor surface. Check for contamination on the motor surface.	Clean the motor surface.
	Overload occurs.	Check the load rate of the drive.	Reduce the load. Increase acceleration/deceleration time. Replace with a larger capacity motor.
	Magnetic force of the magnet has decreased	Check the back EMF and voltage waveform.	Replace the motor.
Abnormal noise occurs	Coupling failure	Check the tightness of the coupling screws and the Check the tightness of the coupling screws and the concentricity of the connection points.	Readjust the coupling.
	Check for bearing abnormalities	Check for bearing vibration and abnormal noise.	Request technical support
	Menu misconfiguration (inertia ratio, gear, filter correction factor)	Check the menu.	Refer to parameter setting method

5.2.2 Subdrive

When an alarm occurs, the fault signal output contact (ALARM) turns OFF, and the motor stops via dynamic braking.

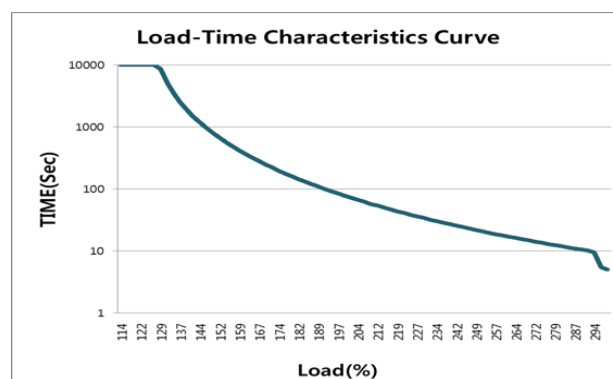
[Alarm Codes and Inspection]

CODE	Description	Description or Cause	Check Item
AL-02	POWER FAIL	Main power is off	Check power supply wiring
AL-03	TTL ENCODER FAIL	TTL encoder phase loss	Check encoder settings, U, V, W wiring, Check connector connection, Replace motor
AL-04	MOTOR OUTPUT FAIL	Motor Shaft Drive Circuit Output Abnormal	Check U, V, W wiring; Check inverter module
AL-05	ENCODER PULSE ERROR	Encoder Pulse Count Setting Abnormal	Calibrate encoder pulse count, replace motor
AL-06	DEVIATION	Pulse loss during position control	Verify position control parameters
AL-07	IGBT OVERHEAT	IGBT Overheating	Check load status, install heat sink,
AL-08	IGBT FAULT	IGBT Drive Error	Wiring, motor, phase resistance, encoder setting values and wiring, Drive replacement
AL-09	OVERLOAD	Overload Alarm	Load status, brake operation status, wiring, motor, Encoder setting value check
AL-10	OVER VOLTAGE	Overvoltage	Input voltage, damaged braking resistor and wiring, excessive regenerative operation
AL-11	OVER SPEED	Over Speed	Encoder malfunction, encoder setting value, encoder wiring, Gain setting, Motor replacement
AL-18	CT-U PHASE OFFSET	Current Sensor U-Phase Offset Error	Current sensor malfunction, drive replacement
AL-19	CT-V PHASE OFFSET	Current Sensor V-Phase Offset Error	Current Sensor Fault, Drive Replacement
AL-20	EEPROM MEMORY	DSP EEPROM Memory Fault	Memory fault, replace drive
AL-21	DRIVE OVERHEAT	Sub-drive high temperature error	Check load, check gain
AL-22	MOTOR OVERHEAT	Motor Overheating	Check load, check gain, Verify acceleration/deceleration correction factor, Verify motor capacity

AL-26	OVER RB	Excessive Regenerative Braking Alarm	Input voltage, regenerative braking resistor and wiring check, Replace drive.
AL-32	PARAM CHANGE REBOOT	Parameter Change Reboot Alarm	Subdrive power re-energized.
AL-35	CT-U PHASE SHORT ERROR	Current Sensor U-Phase Short	Driver replacement
AL-36	CT-U PHASE OVER	Current Sensor U-Phase Oversensing	Check load amount, check gain, replace driver
AL-37	CT-V PHASE SHORT ERROR	Current Sensor V-Phase Short	Driver Replacement

AL-38	CT-V PHASE OVER	Current Sensor V-Phase Oversensing	Check load amount, check gain, replace driver
AL-40	Serial Encoder CRC Error	Serial Encoder Data Error	Check encoder ground wire, check wiring
AL-41	SERIAL ENCODER COM ERROR	Serial Encoder Communication Error	Check encoder ground wire, check wiring
Err1	Error1		Menu that cannot be modified while servo is on To modify, turn off servo first
Error 2	Error2		Enter the data value within the setpoint range
Err3	Error3		Modify [CH809] to Unlock state

5.2.3 Servo Drive Overload Characteristic Curve



Chapter 6

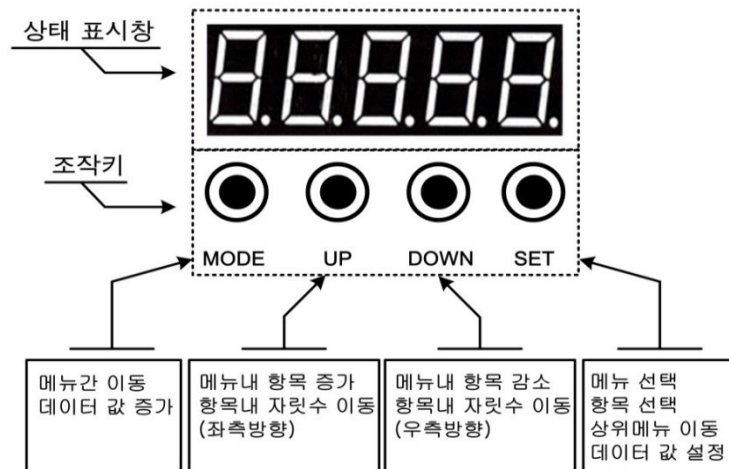
Servo Parameter Settings

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6.1 Status Display Window and Operation Keys



6.2 Status Display Mode Types

Mode Name	Display	Function
Status Display Mode	dr-off	Servo Off Status Display
	dr-on	Servo On Status Display
	AL-00 알람 표시 알람 번호	Display the corresponding alarm when an alarm occurs
Alarm History Check Mode	AL 100 알람 표시 알람이력 번호	Check alarm history
Menu display mode	Pd200 메뉴 표시 항목 번호	Display each menu item
Parameter data setting mode	000000 데이터 설정 값	Display and input parameter data values

6.3 How to Switch Modes



6.4 Parameter Setting Method

Acceleration time [PU608] = 500 [ms] setting example



6.5 Setting the Operating Status Display Parameter

Set the parameters to be displayed on the display window when power is applied to the servo drive.

dP000	Current State	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Current Operating Status	-	~	-	

Displays the driving status of the current drive. The detailed display items are as shown in the following table.

Driving Item	Status Display	Remarks
Servo-On (Servo On)	dr - on	Displayed when the drive is in servo-on state
Servo-Off (Servo Off)	dr - off	Displayed when the drive is in servo-off state
Alarm (Alarm)	AL - □□	When an alarm occurs, the corresponding alarm number is displayed

dP001	Current Speed	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Current Operating Speed	r/min	-10000 to 10000	-	
dP002	Command Speed	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Input Command Speed	r/min	-10000 to 10000	-	
dP003	Not Used	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Disabled	-	-	-	
dP004	Not Used	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Disabled	-	-	-	
dP005	Not Used	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Disabled	-	-	-	
dP006	Not Used	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Disabled	-	-	-	
dP007	Command Torque	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Internal Torque Command	%	-300 to 300	-	
dP008	Torque Limit	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Torque Limit	%	-300 to 300	-	
dP009	Current Load	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Current Load Rate	%	0 ~ 300	-	

dP010	Average Load	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Average Load Factor	%	0 ~ 300	-	
dP011		Unit	Setting Range	Factory Default	Speed Torque Position Control
		-	-	-	
dP012		Unit	Setting Range	Factory Default	Speed Torque Position Control
		-	-	-	
dP013		Unit	Setting Range	Factory Default	Speed Torque Position Control
		-	-	-	
dP014		Unit	Setting Range	Factory Default	Speed Torque Position Control
		-	-	-	
dP015	I/O Status	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Input/Output Contact Status	-	0 ~ 99999	-	

6.6 System Variable Parameter Settings

* Pd200	Motor ID	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Motor ID	-	0 ~ 9	0	

Set the ID of the motor being used.

ID	Model	ID	Model
0	Individual	5	SM□30□□
1	SM□04□□	6	SM□55□□
2	SM□08□□	7	SM□110□□
3	SM□13□□	8	
4	SM□20□□	9	

Pd201	Main Voltage Type	Unit	Setting Range	Shipment Setting	Speed Torque Position Control
	Main Power Supply Rated Voltage	-	0 ~ 1	0	

Select the main power rated voltage for the servo drive.

Pd201	Main power supply rated voltage
0	220V
1	440V

Pd202	U Current Offset	Unit	Setting Range	Factory Default	Speed Torque Position Control
	U-phase Current Offset Setting	mA	-500 to 500	0	
Pd203	V Current Offset	Unit	Setting Range	Factory Default	Speed Torque Position Control
	V-Phase Current Offset Setting	mA	-500 to 500	0	

Sets the current sensor offset value for the servo drive.

Pd204	CCW Torque Limit	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Forward Torque Limit	%	1 ~ 300	300	
Pd205	CW TRQ LIM	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Reverse Torque Limit	%	1 ~ 300	300	

Limits torque to the set value during forward/reverse rotation of the motor. [100%: Motor rated torque]

Pd208	Start Menu Display	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Startup Display Menu	-	1 ~ 19	1	

Sets the operating status parameter number to be displayed on the status display window when power is applied to the servo drive.

Pd209	Inertia	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Inertia	$\text{gf} \cdot \text{cm} \cdot \text{s}^2$	0.01 to 99.99	Motor Model Specific	
Pd210	Torque Constant	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Torque Constant	$\text{kgf} \cdot \text{cm}/\text{A}$	0.01 to 99.99	Motor Model Specific	
Pd211	Phase Inductance	Unit	Setting Range	Factory Default	Speed Torque

				Setting	Position Control
	Phase Inductance	mH	0.1 to 9.999	Motor Model	
Pd212	Phase Resistance	Unit	Setting Range	Factory Default	Speed Torque
	Phase Resistance	Ω	0.001 ~ 9.999	By Motor Model	Position Control
Pd213	Rated Current	Unit	Setting Range	Factory Setting	Speed Torque
	Rated Current	A(rms)	0.01 to 99.99	Motor Model	Position Control
Pd214	MAX Speed	Unit	Setting Range	Factory Default	Speed Torque
	Maximum Speed	r/min	1 to 6000	Motor Model	Position Control
Pd215	Rated Speed	Unit	Setting Range	Factory Setting	Speed Torque
	Rated Speed	r/min	1 to 6000	Motor Model	Position Control
Pd216	Pole Number	Unit	Setting Range	Factory Default	Speed Torque
	Number of Poles	Pole	2 to 16	1	Position Control

Motor parameters between [Pd209] and [Pd216] can only be modified when the motor ID [Pd200] value is '0'.

Pd217	Gain Group	Unit	Setting Range	Factory Default	Speed Torque
	Gain Group	-	0 ~ 9	1	Position Control

** Pd218	Encoder Type	Unit	Setting Range	Factory Default	Speed Torque
	Encoder Type	-	0 ~ 5	1	Position Control

After modifying the parameters, a servo drive reset is required.

Pd218	0	1	2	3	4
Encoder Types	Tamakawa INC	Tamakawa ABS	TTL	Sanyo	Heidenhain

** Pd219	Encoder Pulse	Unit	Setting Range	Factory Setting	Speed Torque
	Encoder Pulse Count	ppr	1 to 32768	32768	Position Control

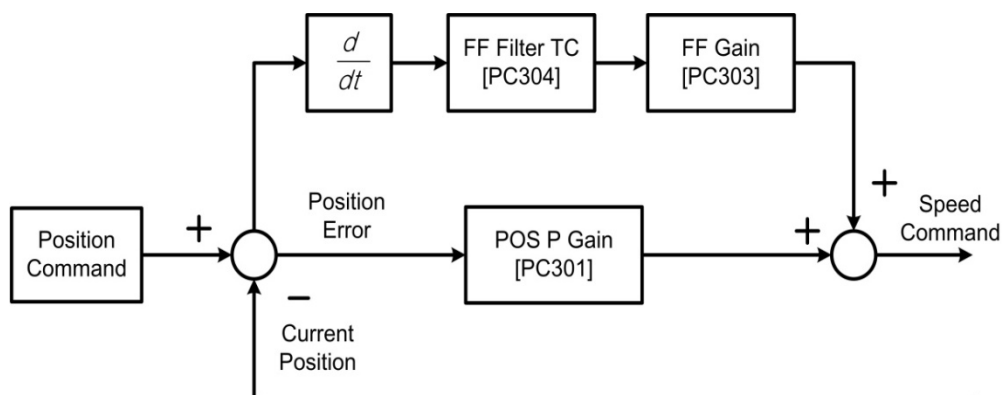
A servo drive reset is required after modifying the parameters.

6.7 System Control Parameter Settings

PC300	Inertia Ratio	Unit	Setting Range	Factory Default	Speed Torque
	System Inertia Ratio	-	1 to 50	1	Position Control

Calculate the ratio of the load inertia of the machine system to the rotor inertia of the motor characteristics and set it as the inertia ratio.

PC301	Position P-Gain 1	Unit	Setting Range	Factory Default	Position Control
	Position Proportional Gain 1	-	1 ~ 500	50	
PC302	Position P-Gain 2	Unit	Setting Range	Factory Default	Position Control
	Position Proportional Gain 2	-	1 ~ 500	70	
PC303	Position FF Rate	Unit	Setting Range	Factory Default	Position Control
	Feed Forward Ratio	%	0 ~ 100	0	
PC304	Position FF Filter TC	Unit	Setting Range	Factory Default	Position Control
	Feedforward Constant	ms	0 to 9999	50	

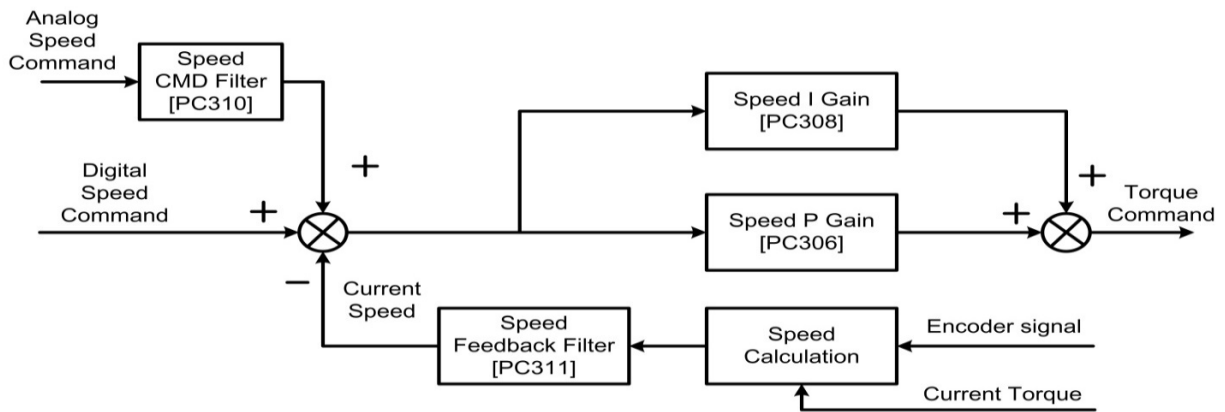


■ [PC301] Position Proportional Gain 1: Multiplies the difference between the position command and the current position by the position proportional gain to convert it into a velocity command.

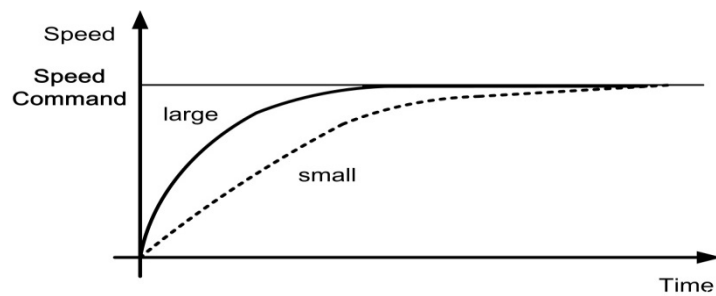
■ [PC303] Feedforward Time Constant: Before using the derivative of the position command as feedforward compensation input, it passes through a filter. This filter's time constant can be set in [ms]. Use a larger time constant value where the position command changes abruptly, and a smaller value where it changes gradually.

■ [PC304] Feedforward Ratio: This adds a velocity command based on the slope of the position command to shorten the position settling time. Setting it too high may cause overshoot and destabilize position control. If this value is '0', the position controller operates in a purely proportional control mode.

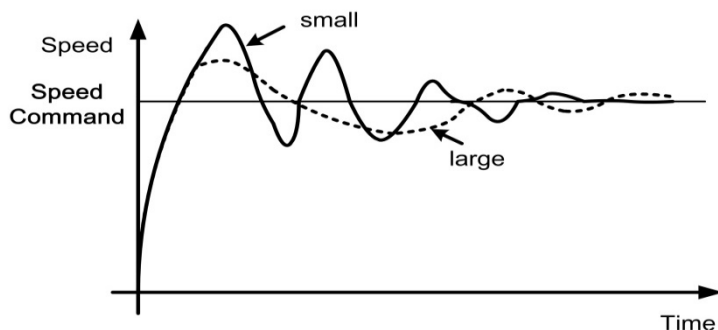
PC305	Position Zero Pulse 1	Unit	Setting Range	Factory Default	Position Control
	Zero Position Pulse 1	10pulse	1 ~ 5000	10	
PC306	Speed P-Gain 1	Unit	Setting Range	Factory Default	Speed Torque Control
	Speed Proportional Gain 1	rad/s	1 ~ 5000	400	
PC307	Speed P-Gain 2	Unit	Setting Range	Factory Default	Speed Torque Control
	Speed Proportional Gain 2	rad/s	1 ~ 5000	400	
PC308	Speed I TC 1	Unit	Setting Range	Factory Default	Speed Position Control
	Speed Integral Time Constant 1	ms	1 to 1000	25	
PC309	Speed I TC 2	Unit	Setting Range	Factory Default	Speed Position Control
	Speed Integral Time Constant 2	ms	1 to 1000	25	
PC310	Speed CMD TC	Unit	Setting Range	Factory Default	Speed Position Control
	Speed Command Correction Factor	ms	0.1 to 99.99	0.50	
PC311	Speed Feedback TC	Unit	Setting Range	Factory Default	Speed Position Control
	Speed Feedback Correction Factor	ms	0.1 to 99.99	0.50	
PC313	Torque CMD TC	Unit	Setting Range	Factory Default	Speed Position Control
	Torque Command Correction Factor	ms	0.1 to 99.99	0.50	



■ [PC306] Speed Proportional Gain 1: Multiplies the difference between the speed command and the current speed by the speed proportional gain to convert it into a torque command. A larger speed proportional gain increases speed response, improving speed tracking but potentially causing vibration. Conversely, a smaller value slows speed response and reduces servo strength.



■ [PC308] Speed Integral Coefficient 1: Calculates the integral value of the difference between the speed command and the current speed, multiplies it by the integral gain, and converts it into a torque command.



■ [PC311] Speed Feedback Correction Constant: This filter suppresses vibration when motor speed fluctuates due to system vibration or when vibration occurs due to gain when a relatively large inertia load is applied. Applying too large a value degrades speed response and reduces control performance.

PC317	COM Mode	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Communication Mode	-	0 ~ 1	0	

PC317	Communication Mode
0	EtherCAT
1	Analog

PC318	Position Zero Pulse 2	Unit	Setting Range	Factory Default	Position Control
	Zero Position Gain 2	10pulse	1 ~ 5000	20	

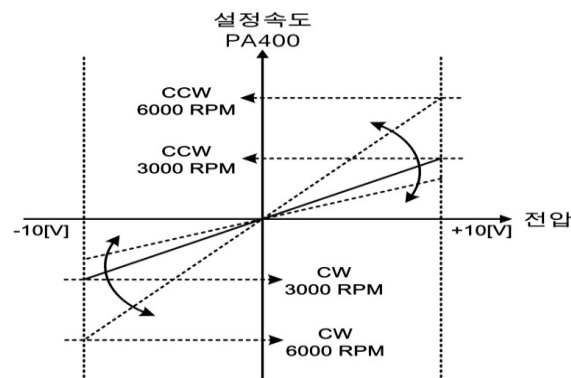
PC319	Zero Speed Lock	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Zero Speed Lock	-	0 ~ 1	0	

6.8 Analog Control Variable Parameter Settings

* PA400	±10[V] Analog SPD	Unit	Setting Range	Factory Default	Speed Torque Control
	±10[V] Analog Speed	r/min	1 ~ 6000	3000	

Cannot be changed while in Servo-On state.

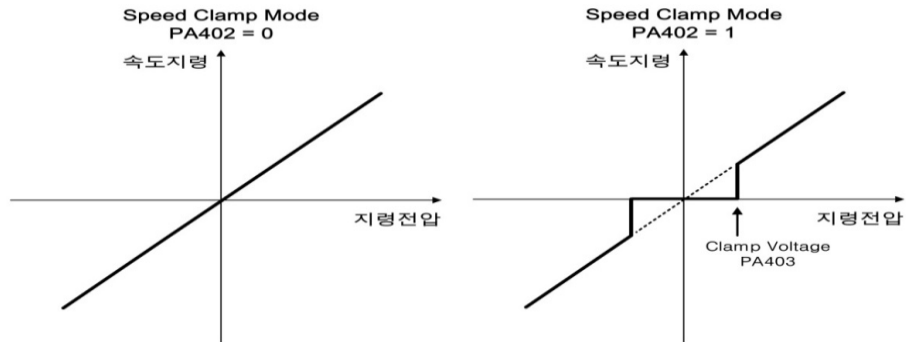
In the servo drive's speed/torque control mode, set the range of command speeds corresponding to ±10[V] analog voltage.



PA401	Speed Offset	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed Command Offset	mV	-1000 to 1000	0	

Sets the offset voltage of the servo drive for the analog speed 0[V] input.

PA402	SPD Clamp Mode	Unit	Setting Range	Shipment Setting Value	Speed Control
	Speed Clamp Mode	-	0 ~ 1	0	
PA403	SPD Clamp Voltage	Unit	Setting Range	Factory Default	Speed Control
	Speed Clamp Voltage	mV	1 to 1000	1	

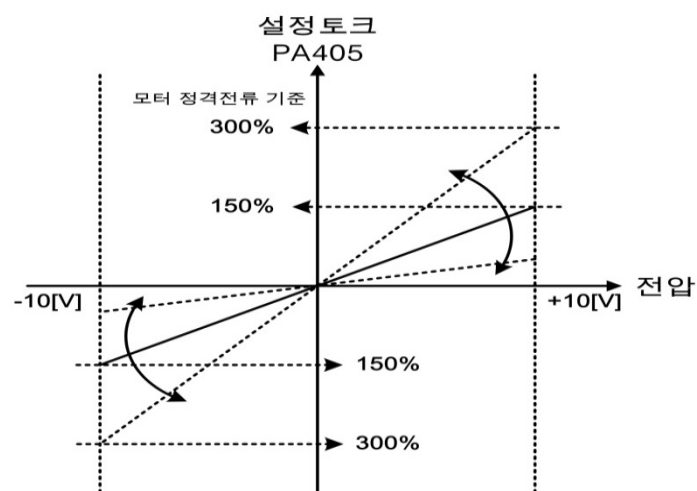


PA404	Speed Override	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed Override	-	0 ~ 1	0	

You can operate with speed command by superimposing a digital speed command over an analog speed command.

Override Mode	Not in operation	Override Operation
PA404	0	1

PA405	±10[V] Analog TRQ	Unit	Setting Range	Factory Default	Speed Torque Position Control
	±10[V Analog Torque	%	1 ~ 300	100	

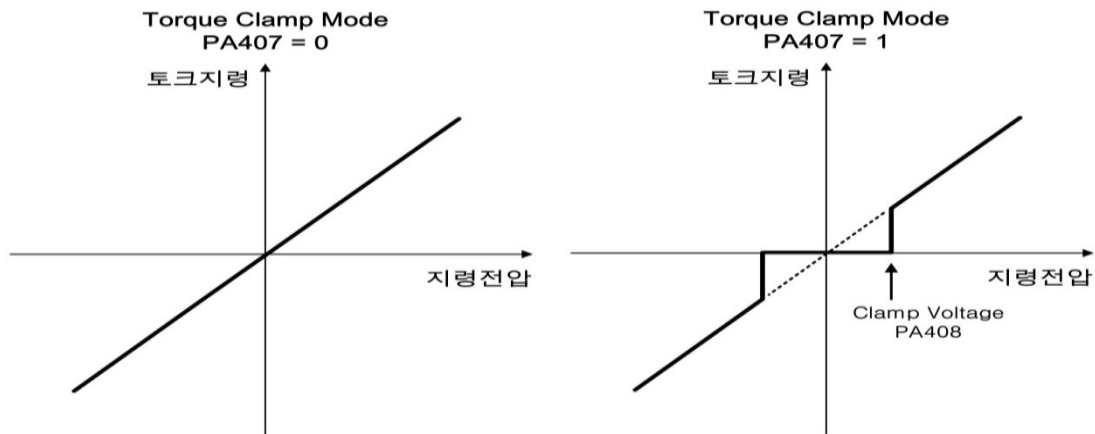


PA406	Torque Offset	Unit	Setting Range	Factory Default	Speed Torque
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	Torque Command Offset	mV	-1000 to 1000	0	Position Control
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Sets the servo drive offset voltage for the analog torque 0[V] input.

PA407	TRQ Clamp Mode	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Torque Clamp Mode	-	0 ~ 1	0	
PA408	TRQ Clamp Voltage	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Torque Clamp Voltage	mV	1 to 1000	0	



PA409	Monitor Type 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Format 1	-	0 ~ 5	1	
PA410	Monitor Mode 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Mode 1	%	0 ~ 1	0	
PA411	Monitor Scale 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Scale 1	%	0.1 ~ 50	1	
PA412	Monitor Offset 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Offset 1	-	0 ~ 5	1	
PA413	Monitor Type 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Format	-	0 ~ 5	1	

	2				
PA414	Monitor Mode 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Mode 2	%	0 ~ 1	0	
PA415	Monitor Scale 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Scale 2	%	0.1 ~ 50	1	
PA416	Monitor Offset 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Monitor Output Offset 2	-	0 ~ 5	1	
PA417	Torque CMD DIR	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Torque Command Direction Setting	-	0 ~ 1	0	

Setting the PA417 parameter value to 1 reverses the torque command direction.

PA418	Reg Brake Time	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Regenerative Braking Operation Time	ms	100 to 1000	500	

When the main power supply voltage exceeds the reference voltage, regenerative braking operates for the set duration to adjust the voltage to the reference value.

PA419	SPD/TRQ Auto Offset	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed/Torque Offset Auto-Tuning Mode	-	0 to 2	0	

This mode automatically adjusts the servo drive's offset voltage for the analog speed/torque command 0[V] input.

After applying a voltage corresponding to zero speed/zero torque to the input pin of the analog input connector (CN6), setting PA419 mode automatically adjusts the voltage applied to the corresponding pin as the offset. Once offset voltage adjustment is complete, it is automatically updated to the offset storage parameter, and the PA419 parameter is released.

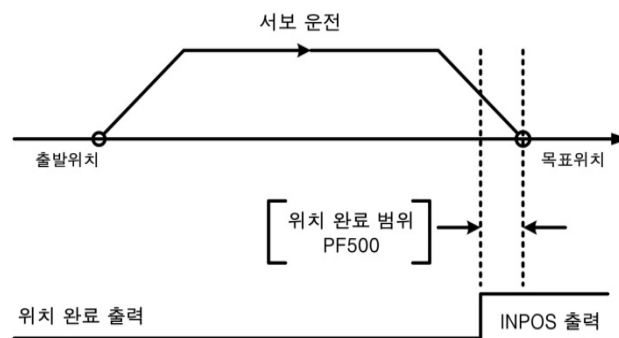
Offset Adjustment Mode	Release	Speed	Torque
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PA419	0	1	2
Offset Storage Location	-	PA401	PA406

6.9 Input/Output Variable Parameter Settings

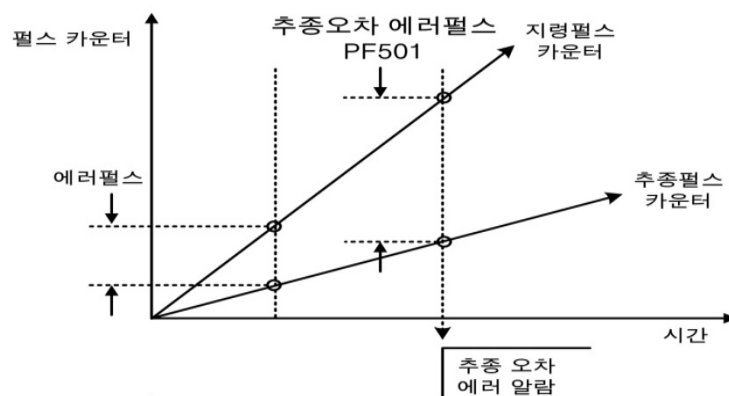
PF500	In Position	Unit	Setting Range	Shipment Setting	Position Control
	Position Completion Range	10pulse	1 ~ 9999	10	

During position control operation, when the target position is reached, a position complete output signal is generated within the set range.



PF501	Follow Error	Unit	Setting Range	Factory Default	Position Control
	Follow Error Pulse	100pulse	1 ~ 9999	900	

A tracking error alarm occurs when the error pulse is greater than the set tracking error pulse.

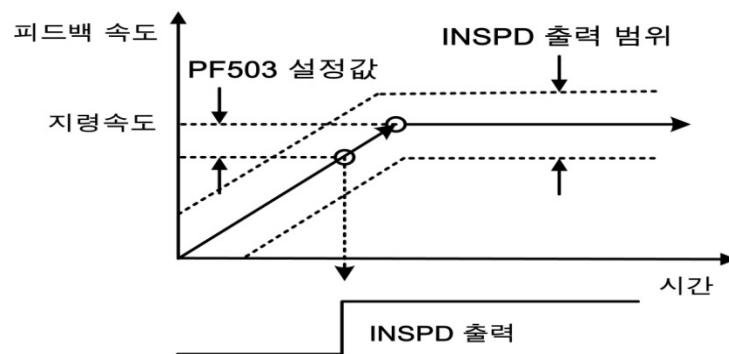


PF502	Zero Speed Range	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Zero Speed Range	r/min	1 to 6000	10	

If the feedback speed is within the set steady speed range, the servo drive recognizes it as steady speed.

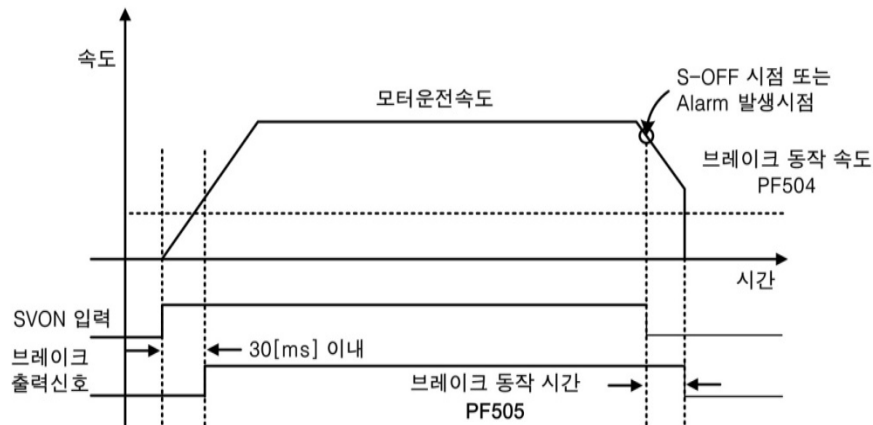
PF503	In Speed	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed Completion Range	r/min	1 to 500	100	

When the target speed is reached during operation, the speed completion output signal (INSPD) is output within the set range.



PF504	Brake Speed	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Brake Operation Speed	r/min	1 to 6000	50	
PF505	Brake Time	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Brake operation time	ms	1 to 1000	10	

The brake output signal activates when either the brake operation speed or operation time requirement is met.



PF506	Power Fail Mode	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Main Power Abnormal Mode	-	0 ~ 1	1	

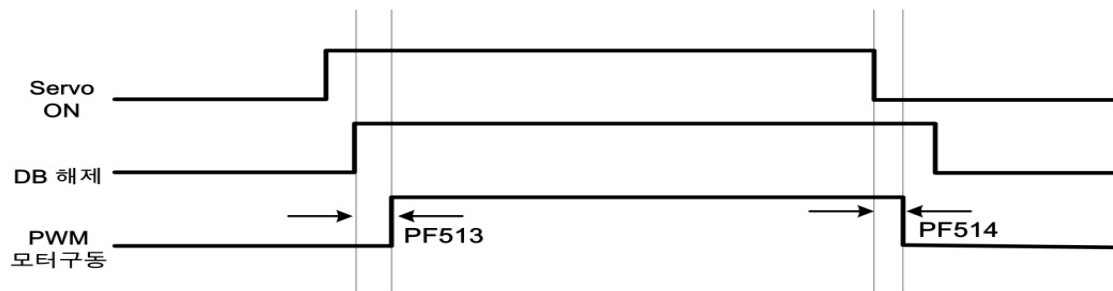
Sets the reset operation mode when an error occurs due to a main power abnormality.

Operation Mode	Disable	Auto Reset
PF506	0	1

PF507	Ready Mode	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Ready Contact Mode Setting	-	0 ~ 1	0	
PF508	Pulse Clear Type	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Pulse Clear Type	-	0 ~ 1	1	
** PF509	Pulse Output Number	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Encoder Resolution	-	1 to 9999	1	
** PF510	Pulse Output Den	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Encoder Resolution Divisor	-	1 to 9999	1	
** PF511	A/B Phase Lead	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Phase Sequence Change for A/B Phases	-	0 ~ 1	1	
** PF512	DIR Select Mode	Unit	Setting Range	Factory Default	Speed Torque

	Speed Operation Direction Setting	-	0 ~ 1	0	Position Control
PF513	PWM On Delay Time	Unit	Setting Range	Factory Default	Speed Torque
	PWM On Delay Time	ms	0 ~ 5000	25	Position Control
PF514	PWM Off Delay Time	Unit	Setting Range	Factory Default	Speed Torque
	PWM Off Delay Time	ms	0 ~ 5000	0	Position Control

Sets the PWM drive ON/OFF delay time.



* PF515	Output Logic 0~2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Output Contact 0~2	-	000 ~111	001	
* PF516	Input Logic 0~3	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Input Contact 0~3	-	0000 ~ 1111	1100	
* PF517	Input Logic 4~7	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Input Contacts 4~7	-	0000 ~ 1111	0000	
PF518	INSPD/Zero SPD	Unit	Setting Range	Factory Default	Speed Torque Position Control
	INSPD/Zero Speed Output	-	0 ~ 1	1	

Output Selection	Zero Speed	In Speed
PF518	0	1

* PF519	DB Control Mode	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Generator Braking Control Mode	-	0 ~ 3	0	

A dynamic brake is built-in to abruptly stop the servo motor during an emergency stop.

The dynamic brake operation mode settings during servo off or stop are as follows.

PF519	Operation Description
0	Dynamic brake decelerates and remains active during servo off (default)
1	When servo is off, decelerate using dynamic brake and perform freewheeling operation below the set zero speed
2	Decelerates to a free-running state when servo is off and maintains the free-running state
3	Decelerate from servo off to free-run state, maintaining dynamic braking below the set zero speed

6.10 Speed Control Parameter Settings

* PU600	Operation Mode	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Operation Mode Selection	-	0 ~ 5	1	

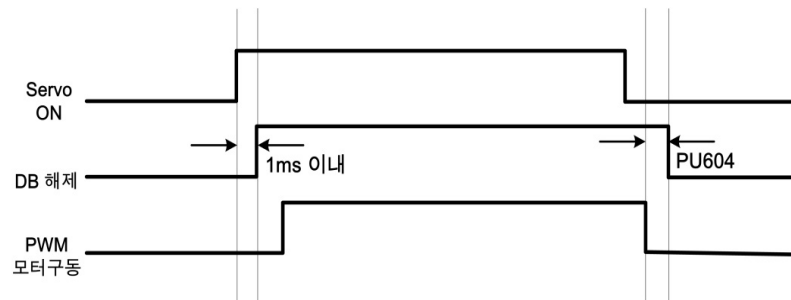
PU600	Driving Mode
0	Torque Mode
1	Speed Mode
2	Position Mode
3	Speed/Location Mode
4	Speed/Torque Mode
5	Position/Torque Mode

PU601	Speed Command 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Internal Speed Command 1	r/min	-6000 to 6000	200	
PU602	Speed Command 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Internal Speed Command 2	r/min	-6000 to 6000	1000	
PU603	Speed Command 3	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Internal Speed Command 3	r/min	-6000 to 6000	3000	

Internal Digital Speed Command This is used as a speed override mode and as a speed command for manual jog operation [CH802] via the setting parameter.

PU604	DB Delay Time	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Generator Braking Delay Time	ms	0 ~ 2000	1	

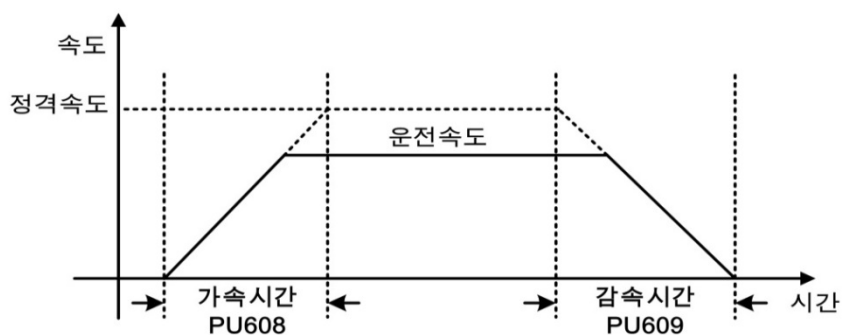
Sets the operation delay time of the dynamic brake.



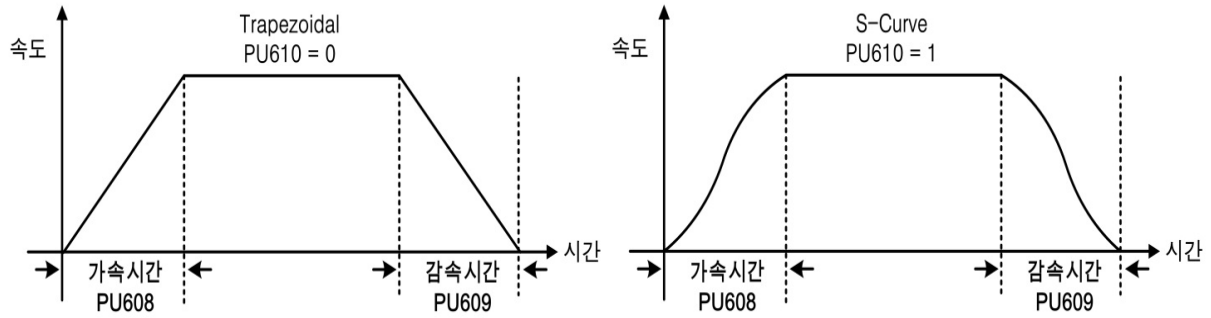
PU606	Z Phase Search SPD	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Z-axis Search Drive Speed	r/min	1 to 300	10	
PU607	Manual Position SPD	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Manual Position Drive Speed	r/min	1 to 6000	500	
PU608	Accel Time	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed Command Acceleration Time	ms	0 to 9999	0	
PU609	Deccel Time	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed Command Deceleration Time	ms	0 ~ 9999	0	

Definition of Acceleration/Deceleration Time

- Acceleration time: The time required to accelerate from a stationary state to the motor's rated speed.
- Deceleration time: The time required to decelerate from the motor's rated speed to a stop.



* PU610	S-Curve Control	Unit	Setting Range	Factory Default	Speed Torque Position Control
	S-curve operation	-	0 ~ 1	0	



PU611	JOG Speed 0	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Speed 0	r/min	-6000 to 6000	100	
PU612	Jog Speed 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Speed 1	r/min	-6000 to 6000	100	
PU613	Jog Speed 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Speed 2	r/min	-6000 to 6000	100	
PU614	Jog Speed 3	Unit	Setting Range	Shipment Setting Value	Speed Torque Position Control
	Jog Speed 3	r/min	-6000 to 6000	100	
PU615	Jog Time 0	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Time 0	Sec	1 ~ 3600	5	
PU616	Jog Time 1	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Time 1	Sec	1 ~ 3600	5	
PU617	Jog Time 2	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Time 2	Sec	1 ~ 3600	5	
PU618	JOG Time 3	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Jog Time 3	Sec	1 ~ 3600	5	

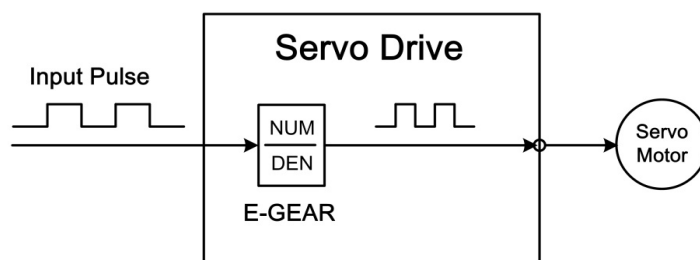
This parameter is used to set the speed command and operation time during Auto Jog operation. Refer to the Auto Jog Operation [CH803] parameter section.

PU619	Not Used	Unit	Setting Range	Factory Default	
	Not Used	-	-	-	

6.11 Position Drive Parameter Settings

PP700	POS ACC	Unit	Setting Range	Factory Default	Position Control
	Position Control Acceleration Time	ms	1 to 1000	1	
PP701	Electric Gear Number 0	Unit	Setting Range	Factory Setting	Position Control
	Electromagnetic Gear Ratio Numerator 0	-	1 ~ 9999	1000	
PP702	Elec Gear Den0	Unit	Setting Range	Shipment Setting	Position Control
	Electromagnetic Gear Ratio Denominator 0	-	1 ~ 9999	1000	
PP703	Electrical Gear Number 1	Unit	Setting Range	Factory Setting	Position Control
	Electromagnetic Gear Ratio Numerator 1	-	1 ~ 9999	1000	
PP704	Electrical Gear Den1	Unit	Setting Range	Shipment Setting	Position Control
	Electromagnetic Gear Ratio Denominator 1	-	1 ~ 9999	1000	
PP705	Electrical Gear No. 2	Unit	Setting Range	Factory Setting	Position Control
	Electromagnetic Gear Ratio Numerator 2	-	1 ~ 9999	1000	
PP706	Electrical Gear Den2	Unit	Setting Range	Shipment Setting	Position Control
	Electromagnetic Gear Ratio Denominator 2	-	1 ~ 9999	1000	
PP707	Electrical Gear Number 3	Unit	Setting Range	Shipment Setting	Position Control
	Electromagnetic Gear Ratio Numerator 3	-	1 ~ 9999	1000	
PP708	Elec Gear Den3	Unit	Setting Range	Shipment Setting	Position Control
	Electromagnetic Gear Ratio Denominator 3	-	1 ~ 9999	1000	

Electronic Gear Ratio Setting



$$\text{Electronic Gear Ratio} \left(\frac{\text{분자}}{\text{분모}} \right) = \frac{\text{엔코더 펄스 수} \times 4}{\text{지령단위}} \times \text{Reduction Ratio} \left(\frac{b}{a} \right)$$

- Reduction Ratio: Indicates that the load shaft rotates b times for every a rotations of the motor
- Command Unit: The smallest unit of position data required to move the load
- Encoder Pulse Count

Encoder Type	Encoder Pulse Count (ppr)
Incremental Encoder	2000 ~ 6000
Serial 17-bit	32768

PP709	Not Used	Unit	Setting Range	Factory Default	
	Not Used				
PP710	Not Used	Unit	Setting Range	Shipment Setting	
	Not Used				
PP711	Not Used	Unit	Setting Range	Shipment Setting	
	Not Used				
PP712	Not Used	Unit	Setting Range	Shipment Setting	
	Not Used				
PP713	Not Used	Unit	Setting Range	Shipment Setting	
	Not Used				
PP714	Not Used	Unit	Setting Range	Shipment Setting	
	Not Used				
PP715	Not Used	Unit	Setting Range	Shipment Setting	
	Not Used				
PP716	Absolute Singleton Subunit	Unit	Setting Range	Shipment Setting Value	Speed Torque Position Control
	ABS Single Turn Low	0	0 ~ 65535	-	
PP717	Absolute Single Turn Upper	Unit	Setting Range	Shipment Setting Value	Speed Torque Position Control
	ABS Single Turn High	0	65535	-	
PP718	Absolute Multi-Turn Sub	Unit	Setting Range	Factory Default	Speed Torque Position
	ABS Multi Turn Low	0	65535	-	

					Control
PP719	Absolute Multi-Turn Upper	Unit	Setting Range	Factory Default	Speed Torque Position
	ABS Multi Turn High	0	65535	-	Control

6.12 Setting Drive Operation Parameters

CH800	Alarm Reset	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Alarm Reset	-	-	-	

When an alarm occurs, selecting the CH800 parameter will clear the alarm.

(Excluding AL-08, AL-32, AL-40)

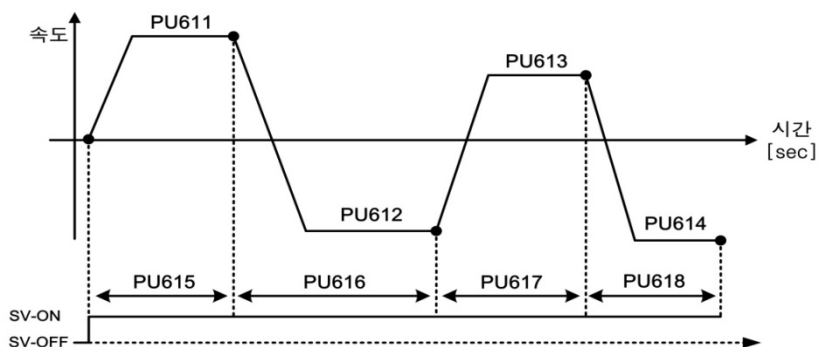
CH801	Alarm History Clear	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Alarm History Clear	-	-	-	

Alarm history stored in AL100~AL119 will be deleted.

CH802	Manual JOG Run	Unit	Setting Range	Factory Default	Speed Control
	Manual Jog Operation	-	-	-	

CH803	Auto JOG Run	Unit	Setting Range	Factory Default	Speed Control
	Auto JOG Run	-	-	-	

Auto jog operation can be performed using the jog speed parameters [PU611]~[PU614] and jog time parameters [PU615]~[PU618].



CH805	Z Phase Search	Unit	Setting Range	Factory Default	Position Control
	Z-Phase Search Operation	-	-	-	
CH806	Manual Position Run	Unit	Setting Range	Shipment Setting Value	Position Control
	Manual Position Operation	-	-	-	
CH807	Output Test	Unit	Setting Range	Shipment Setting	Speed Torque Position Control
	Output Contact Test	-	-	-	

The three output contacts can be forcibly tested ON/OFF using the operation key.

CH808	Menu Data Init	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Menu Reset	-	-	-	
CH809	Menu Data Lock	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Menu Lock	-	-	-	
CH810		Unit	Setting Range	Factory Default	Position Control
		-	-	-	
CH811	Current Offset Save	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Current Offset Compensation	-	-	-	
CH812	Speed Offset Save	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Speed Offset Save	-	-	-	
CH813	Torque Offset Save	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Torque Offset Save	-	-	-	
CH814	Maximum Load	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Instantaneous Maximum Load Ratio	%	-1000 to 1000	-	
CH815	Feedback Pulse	Unit	Setting Range	Shipment Setting Value	Position Control

	Follow-up Position Pulse	pulse	0 to 99999	1	
CH819	All Menu Data Init	Unit	Setting Range	Factory Default	Speed Torque Position Control
	Reset All Menu Values	-	-	-	

6.13 Alarm Monitoring Parameters

This parameter allows you to view alarm history and cannot be modified.

Alarm History	Description
AL100	Display recent alarms
AL101	Display previous alarm
AL102	Display alarms from two previous occurrences
⋮	⋮
AL119	Display alarms occurring before the 19th occurrence

- Alarm Code Detailed Description

CODE	Name	Description or Cause	Inspection Item
AL-02	POWER FAIL	Main power is off	Check power supply wiring
AL-03	TTL ENCODER FAIL	TTL encoder phase loss	Verify encoder settings, U, V, W wiring, Check connector connection, Replace motor
AL-04	MOTOR OUTPUT FAIL	Motor shaft drive circuit output abnormality	Check U,V,W wiring, Check inverter module
AL-05	ENCODER PULSE ERROR	Encoder pulse count setting error	Calibrate encoder pulse count, replace motor
AL-06	DEVIATION	Pulse loss during position control	Check position control parameters
AL-07	IGBT OVERHEAT	IGBT Overheating	Check load status, install heat sink,
AL-08	IGBT FAULT	IGBT Drive Error	Wiring, motor, phase resistance, encoder setting values and Wiring, Drive Replacement
AL-09	OVERLOAD	Overload Alarm	Load status, brake operation status, wiring, motor, Encoder setting value check
AL-10	OVER VOLTAGE	Overvoltage	Input voltage, damaged braking resistor and wiring, excessive regenerative operation
AL-11	OVER SPEED	Over Speed	Encoder malfunction, encoder set value, encoder Wiring, Gain Setting, Motor Replacement
AL-18	CT-U PHASE OFFSET	Current Sensor U-Phase Offset Error	Current sensor fault, drive replacement

AL-19	CT-V PHASE OFFSET	Current Sensor V-Phase Offset Error	Current Sensor Fault, Drive Replacement
AL-20	EEPROM MEMORY	DSP EEPROM memory fault	Memory fault, replace drive
AL-21	DRIVE OVERHEAT	Sub-drive high temperature error	Check load, check gain
AL-22	MOTOR OVERHEAT	Motor Overheating	Check load, check gain, Verify acceleration/deceleration correction factor, Verify motor capacity
AL-26	OVER RB	Excessive Regenerative Braking Alarm	Input voltage, regenerative braking resistor and wiring check, drive replacement.

CODE	Description	Details or Cause	Inspection Item
AL-32	PARAMETER CHANGE REBOOT	Parameter Change Reboot Alarm	Subdrive power re-energized.
AL-35	CT-U PHASE SHORT ERROR	Current Sensor U-Phase Short	Driver Replacement
AL-36	CT-U PHASE OVER	Current Sensor U-Phase Oversensing	Check load amount, check gain, replace driver
AL-37	CT-V PHASE SHORT ERROR	Current Sensor V-Phase Short	Driver Replacement
AL-38	CT-V PHASE OVER	Current Sensor V-Phase Oversensing	Check load amount, check gain, replace driver
AL-40	SERIAL ENCODER CRC ERROR	Serial Encoder Data Error	Check encoder ground wire, check wiring
AL-41	SERIAL ENCODER COM ERROR	Serial Encoder Communication Error	Check encoder ground wire, check wiring
Err1	Error1		Menu cannot be modified while servo is on To modify, turn off servo first
Err2	Error2		Within the setpoint range, the corresponding data value Err3
Err3	Error3		Modify [CH809] to Unlock state

Chapter 7

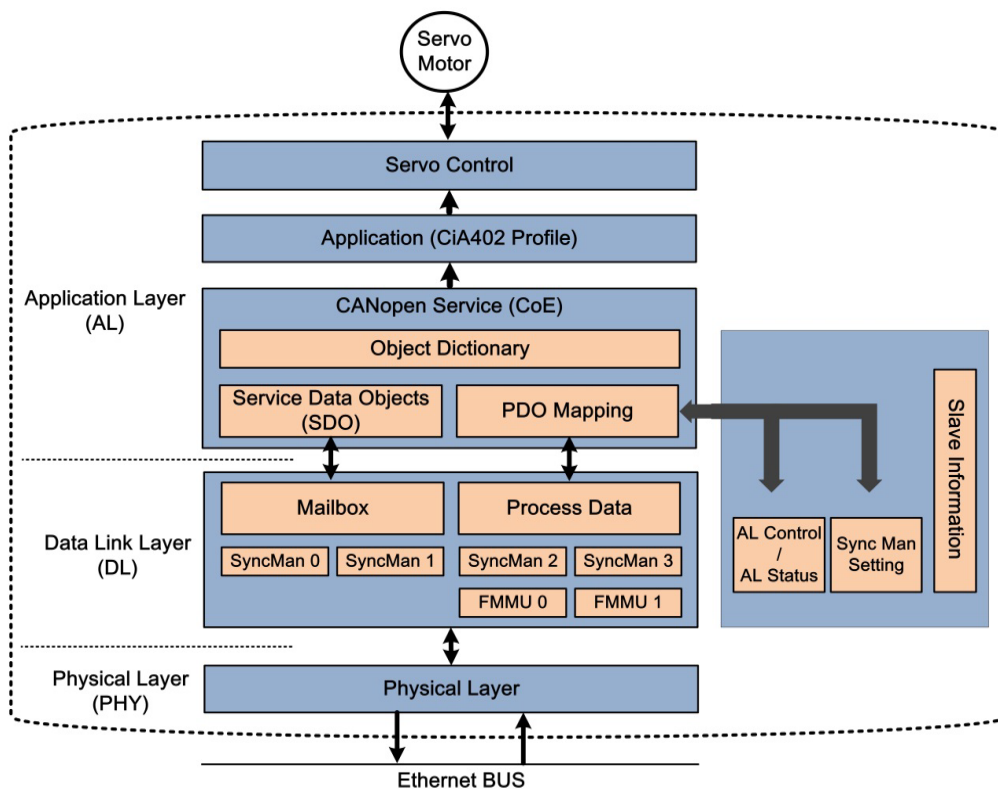
EtherCAT Communication

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EtherCAT stands for Ethernet for Control Automation Technology. It is a communication protocol between masters and slaves using Real-Time Ethernet, developed by BECKHOFF in Germany and managed by the ETG (EtherCAT Technology Group). EtherCAT technology overcomes the inherent limitations of other Ethernet solutions, such as real-time performance and data transfer rates, and interprets, processes, and copies data on all devices. EtherCAT slave devices insert input data while the frame passes through the node, with the frame experiencing only the hardware transmission delay time (ns). EtherCAT uses IEEE 802.3 standard frames, ensuring data is delivered intact to the target device. It supports various topologies, including line, tree, and star. Furthermore, using high-speed Ethernet (100Base TX, built-in network server), the cable length between two devices can reach up to 100 meters, and up to 65,535 devices can be connected, making the network size virtually unlimited.

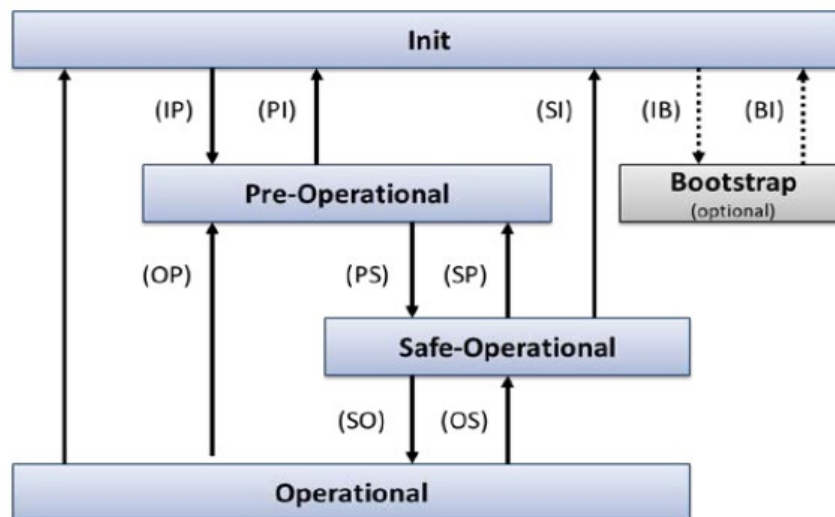
7.1 CoE Architecture

The following diagram illustrates the EtherCAT (CoE) architecture of the servo drive. It utilizes the CiA402 drive profile, and the application layer's object dictionary contains application data, process data interfaces, and PDO mapping information between application data. The PDO (Process Data Object) consists of an object dictionary mappable to the PDO, and the content of the process data is defined by the the PDO mapping. Process data communication reads and writes the PDO periodically, while mailbox communication is non-periodic and can read and write all object dictionaries.



7.2 EtherCAT State Machine (ESM)

The following figure illustrates the EtherCAT state machine.

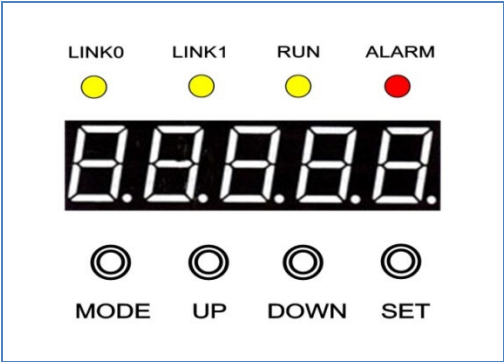


State	Description
INIT	- Initialization state - Mailbox/PDO communication not possible
PREOP	- Mailbox communication possible
SAFEOP	- PDO Input (TxPDO) communication possible
OP	- PDO Input (TxPDO) communication enabled - PDO Output (RxPDO) communication possible

State Transition	Description
IP	- Start mailbox communication
PI	- Stop mailbox communication
PS	- Start input data update
SP	- Stop input data update
SO	- Start output data update
OS	- Stop updating output data
OP	- Stop input/output data update
SI	- Stop input data updates and mailbox communication
OI	- Stop both input/output data updates and mailbox communication

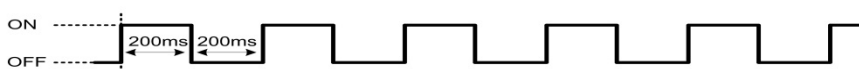
7.3 LED Status

The front panel of the servo drive is equipped with three LEDs conforming to the EtherCAT standard, as shown in the figure below. These LEDs indicate communication status and error conditions, playing a crucial role in diagnosing and resolving network issues.



7.3.1 Link Activity LED

The LINK0 LED indicates the status of the CN6 communication port, while the LINK1 LED indicates the status of the CN7 communication port. The meanings of the LED illumination states are as follows.

Link Activity LED	Description
Off	No communication connection.
Flickering	Communication connection is established and communication is active. 
On	Communication connection is established, but communication is not active.

7.3.2 RUN LED

RUN LED	Description
Off	In Init (INIT) state.
Blinking	In Pre-Operation (PREOP) state. 
Single Flash	Safe-Operation (SAFEOP) state. 
On	Operational (OP) state.

7.3.3 ERROR LED

ERROR LED	Description
Off	EtherCAT communication is in a normal state.
Flickering	Booting error occurred.
Blinking	Received an object configuration command from the master that cannot be set in the current state.
Single Flash	State changed without a master command due to a Sync Error.
Double Flash	A watchdog error occurred during communication.
On	Serious communication-related issue occurred.

7.4 Data Type

Name	Data Type	Range
SINT	Signed 8-bit	-128 to 127
USINT	Unsigned 8-bit	0 ~ 255
INT	Signed 16-bit	-32768 to 32767
UINT	Unsigned 16-bit	0 to 65535
DINT	Signed 32-bit	-21247483648 ~ 21247483647
UDINT	Unsigned 32-bit	0 ~ 4,294,967,295
STRING	String	-

7.5 CoE Message Type

To access the object dictionary, the servo drive provides CoE in two ways. Process Data Objects (PDOs) are organized in the object dictionary defined in the PDO mapping. Process data communication at the Data Link Layer is periodic communication that can write to and read from PDOs, while Service Data Objects (SDOs) are non-periodic communication that can be read from and written to via mailbox communication. SDO Request and SDO Response are provided for accessing the object dictionary.

The CoE message types are as follows.

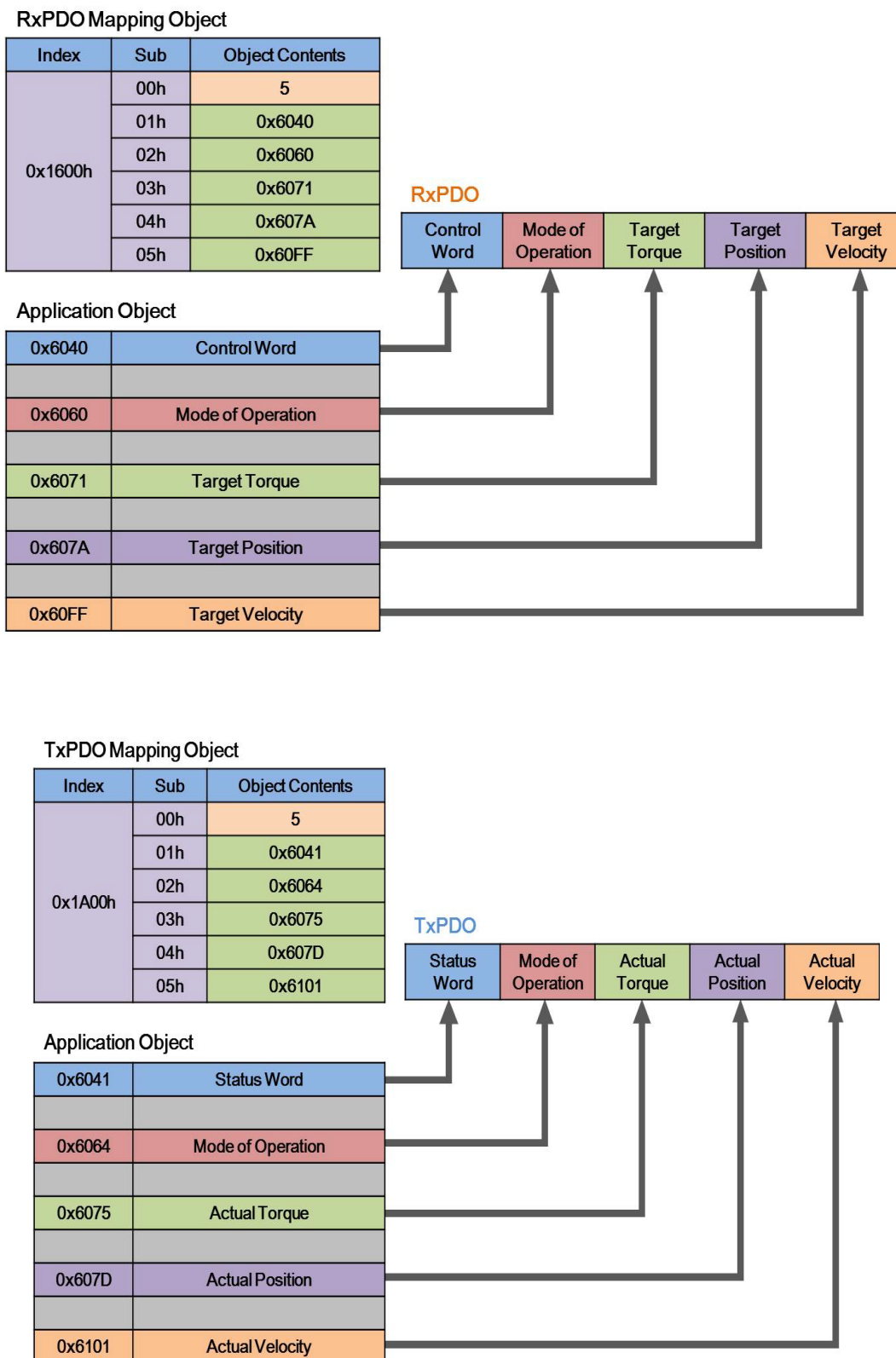
Type Value	Description	Support
2	SDO Request	Support
3	SDO Response	Support
8	SDO Information	Not Supported

7.6 PDO Mapping

Real-time data transmission in EtherCAT is performed via PDO (Process Data Object). PDO includes RxPDO, which receives data transmitted from the master controller, and TxPDO, which transmits status from the drive to the master controller.

The servo drive uses 0x1600 to 0x1603 for RxPDO mapping and 0x1A00 to 0x1A03 for TxPDO mapping. Each PDO can map up to 10 objects.

The following figure shows an example of mapping the 1stRxPDO and 1stTxPDO.



7.7 Distributed Clock

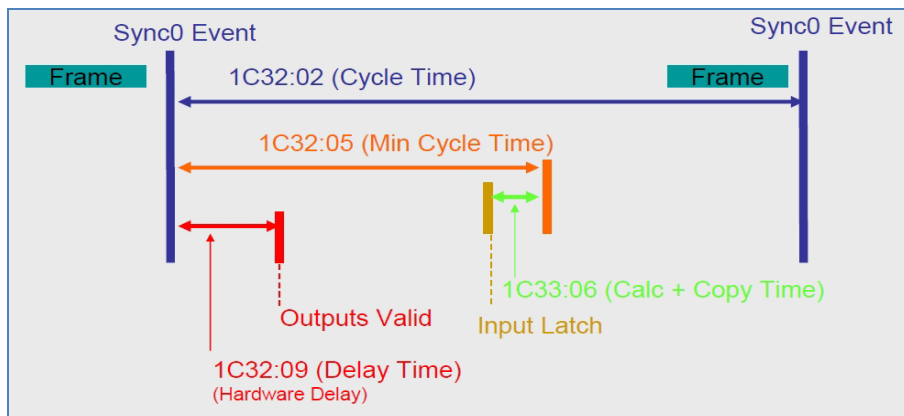
A distributed clock is used for synchronization in EtherCAT communication. The master and slaves synchronize by sharing a Reference Clock (System Time), and the slave synchronizes the application using the Sync0 event generated by the Reference Clock. The synchronization mode can be changed via the Sync Control Register.

■ Free-Run Mode

In Free-Run mode, each device operates on its own cycle, independent of the communication cycle and master cycle.

■ DC Synchronous Mode

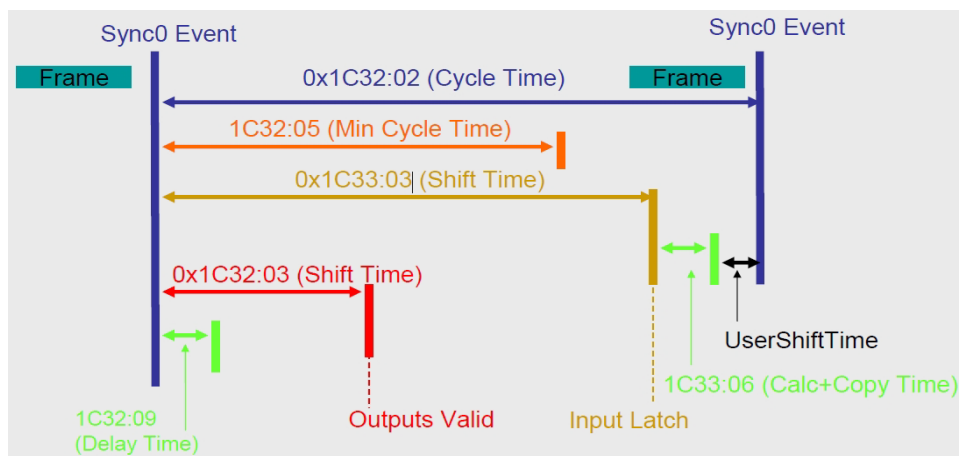
In DC Synchronous mode, the drive (slave) is synchronized by the master's Sync0 event.



The frame is received before the Sync0 event.

Sync0 Cycle Time > 0x1C32:05

0x1C32:01 = 0x1C33:01 = 2, 0x1C32:06 = 0, 0x1C32:09 = 0



Output Shift Time 0x1C32:03 > 0x1C32:06

Input Shift Time 0x1C33:03 > 0x1C32:05 - 0x1C33:06

Input Shift Time

 $0x1C33:03 < \text{Sync0 Cycle Time} - \text{UserShiftTime} - 0x1C33:06$

7.8 Object Dictionary

The Object Dictionary at the Application Layer contains parameters, application data, process data, and mapping information for application data. The servo drive's Object Dictionary implements the CiA402 profile.

7.9 Sync Manager ()

■ SyncManager provides interrupts for consistent and secure data exchange between EtherCAT masters and slaves and is configured by the master. It can configure both the communication mode and direction, and uses buffer locations in memory areas for data exchange. Access to the buffers is controlled by the SyncManager hardware.

SyncManager Channel	Operation Mode	Start Address	Description
0	Receive Mailbox	0x1800	- Used for mailbox communication mode - Non-periodic data input channel - Used for Mailbox communication mode
1	Transmit Mailbox	0x1C00	- Non-periodic data output channel - Used for Process Data communication mode
2	Receive PDO	0x1100	- Periodic Data Input Channel - Process Data communication mode used
3	Transmit PDO	0x1400	- Periodic Data Output Channel

7.10 Sync Manager Communication Mode

■ Mailbox Mode

Mailbox Mode implements a handshake mechanism for data exchange. The EtherCAT master and slave (servo drive) can access the buffer only after the buffer access has been completed. When the

sender writes to the buffer, the buffer remains in lock mode until the receiver completes reading the data, ensuring no data loss. Mailbox Mode is used for application layer protocols and non-periodic data exchange.

Chapter 8

Standard EtherCAT Servo Parameter Settings

- 8.1 CoE Communication Objects.....93
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- 8.4 Manufacturer Specific Objects.....102

This chapter describes how to configure individual servo parameters according to their intended use. Settings for the menu require the use of an EtherCAT master. Among the displayed parameter numbers, parameters marked with an asterisk (*) cannot be modified while the servo is active (Servo-On). Parameters marked with two asterisks (**) require a system reset for the changed values to be reflected in the servo drive.

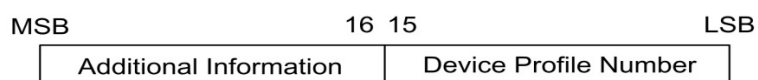
The abbreviations and their meanings used in this manual are as shown in the following table.

Abbreviation	Meaning	
ACC	Acceleration	Acceleration
DEC	Deceleration	Deceleration
CMD	Command	Command
SPD	Speed	Speed
POS	Position	Position
TRQ	Torque	Torque
INIT	Initialize	Initialization
LIM	Limit	Limit
EN	Enable	Allow
FF	Feed Forward	Feed Forward
ERR	Error	Error
NUM	Numerator	Numerator
DEN	Denominator	Denominator
MAX	Maximum	Maximum
MIN	Minimum	Minimum
REV	Revolution	Rotation
TC	Time Constant	Time Constant
CONV	Conversion	Conversion

8.1 CoE Communication Objects

Index		Device Type					
0x1000							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Device Type	UDINT	Read	No	-	0x00020192	-

■ Description



Additional Information: 0x0002 (Servo Drive)

Device Profile Number: 0x0192 (CiA402)

Index		Error Register					
0x1001							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Error Register	USINT	RO	No	-	0x00	-

■ Description

Bit	Function	Value(Hex)	Setting Details
0	General Error	0	No Error
		1	Error
1 to 7	Reserved	-	0 : Always

Index		Manufacturer Device Name					
0x1008							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Manufacturer Device Name	STRING	RO	No	-	0x0	-

Index		Hardware Version					
0x1009							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Hardware Version	STRING	Read Only	No	-	-	-

Index		Software Version					
0x100A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Software Version	STRING	RO	No	-	-	-

Index		Store Parameters					
0x1010							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	UDINT	RW	No	4	0x04	-
1	Store all Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-
2	Store Communication Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-
3	Store CiA402 Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-
4	Store Specific Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-

■ To prevent parameters from being incorrectly stored, the Sub-Index records a specific "save" when storing a parameter.

	MSB	16	15	LSB
ASCII	e	v	a	s
Hex	0x65	0x76	0x61	0x73

- All Parameters are stored when "save" is written to Sub-Index 1.
- Communications are stored when "save" is written to Sub-Index 2.
- CiA402 Parameters are stored when "save" is written to Sub-Index 3.
- Specific Parameters are stored when "save" is written to Sub-Index 4.

Index		Restore Parameters					
0x1011							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	UDINT	RW	No	4	0x04	-
1	Restore Default Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-
2	Restore Communication Default Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-
3	Restore CiA402 Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-
4	Restore Specific Parameters	UDINT	RW	No	0x00000000 ~ 0xFFFFFFFF	0x0	-

■ To prevent parameters from being incorrectly stored, the Sub-Index records a specific "save" when storing a parameter.

	MSB	16	15	LSB
ASCII	d	a	o	l
Hex	0x64	0x61	0x6F	0x6C

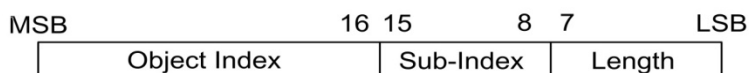
- All Parameters are restored when "load" is written to Sub-Index 1.
- Communications are restored when "load" is written to Sub-Index 2.
- CiA402 Parameters are restored when "load" is written to Sub-Index 3.
- Specific Parameters are restored when "load" is written to Sub-Index 4.
- Turn the power off and then back on to restore the default values.

Index		Identity Object					
0x1018							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RO	No	-	0x04	-
1	Vendor ID	UDINT	RO	No	-		-
2	Product Code	UDINT	RO	No	-	0x00	-
3	Revision Number	UDINT	RO	No	-	0x0	-
4	Serial Number	UDINT	RW	No		0x0	-

8.2 PDO Mapping Objects

Objects can be mapped to Process Data Objects (PDOs) using PDOs for real-time data transmission via the CANopen over EtherCAT protocol.

Used for configuring receive PDO mapping and transmit PDO mapping. Displays information about the mapped application object.



Bit 0-7: Bit length of the mapped Object (e.g., displayed as 0x20 for 32-bit)

Bit 8-15: Sub-Index of the mapped Object

Bit 16-31: Index of the mapped object

0x1600 ~ 0x1603, Receive PDO Mapping

1st Receive PDO Mapping

Index		1 st Receive PDO Mapping					
0x1600							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~8	5	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60400010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x60710010	-
3	Mapping entry 3	UDINT	RW	No	0~0xFFFFFFFF	0x607A0020	-
4	Mapping entry 4	UDINT	RW	No	0~0xFFFFFFFF	0x60600008	-
5	Mapping entry 5	UDINT	RW	No	0~0xFFFFFFFF	0x60B80010	

2ndReceive PDO Mapping

Index		2 nd Receive PDO Mapping					
0x1601							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~8	2	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60400010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x607A0020	-

3rd Receive PDO Mapping

Index		3 rd Receive PDO Mapping					
0x1602							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~8	2	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60400010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x60FF0020	-

4th Receive PDO Mapping

Index		4 th Receive PDO Mapping					
0x1603							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~8	2	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60400010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x60710010	-

0x1A00 ~ 0x1A03, Transmit PDO Mapping

1st Transmit PDO Mapping

Index		1 st Transmit PDO Mapping					
0x1A00							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~10	10	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60410010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x60770010	-
3	Mapping entry 3	UDINT	RW	No	0~0xFFFFFFFF	0x60640020	-
4	Mapping entry 4	UDINT	RW	No	0~0xFFFFFFFF	0x60F40020	-
5	Mapping entry 5	UDINT	RW	No	0~0xFFFFFFFF	0x60FD0020	-
6	Mapping entry 6	UDINT	RW	No	0~0xFFFFFFFF	0x60610008	-
7	Mapping entry 7	UDINT	RW	No	0~0xFFFFFFFF	0x26010010	-
8	Mapping entry 8	UDINT	RW	No	0~0xFFFFFFFF	0x26000010	-
9	Mapping entry 9	UDINT	RW	No	0~0xFFFFFFFF	0x60B90010	
10	Mapping entry 10	UDINT	RW	No	0~0xFFFFFFFF	0x60BA0020	

2nd Transmit PDO Mapping

Index		2 nd Transmit PDO Mapping					
0x1A01							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~10	2	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60400010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x607A0020	-

3rd Receive PDO Mapping

Index		3 rd Transmit PDO Mapping					
0x1A02							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~10	2	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x60410010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x60640020	-

4th Receive PDO Mapping

Index		4 th Transmit PDO Mapping					
0x1A03							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RW	No	0~10	2	-
1	Mapping entry 1	UDINT	RW	No	0~0xFFFFFFFF	0x6041 0010	-
2	Mapping entry 2	UDINT	RW	No	0~0xFFFFFFFF	0x6064 0020	-

8.3 Sync Manager Communication Objects

0x1C00 Sync Manager Communication Type

Index		Sync Manager Communication Type					
0x1C00							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RO	No	-	4	-
1	Communication Type Sync Manager 0	USINT	RO	No	-	1: mailbox receive (Master to Slave)	-
2	Communication Type Sync Manager 1	USINT	RO	No	-	2: mailbox Send (Slave to Master)	-
3	Communication Type Sync Manager 2	USINT	RO	No	-	3: Process Data Output (Master to Slave)	-
4	Communication Type Sync Manager 3	USINT	RO	No		4: Process Data Input (Slave to Master)	-

0x1C10~0x1C13 Sync Manager PDO Assignment

Index		Sync Manager 0 PDO Assignment					
0x1C10							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Sync Manager 0 PDO Assignment	USINT	Read-Only	No	-	-	-

Index		Sync Manager 1 PDO Assignment					
0x1C11							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Sync Manager 1 PDO Assignment	USINT	Read-Only	No	-	-	-

Index		Sync Manager 2 PDO Assignment					
0x1C12							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RO	No	0~2	1	-
0	Index of assigned RxPDO 1	UINT	RW	No	0x1600 ~ 0x1603	0x1601	-

Index		Sync Manager 3 PDO Assignment					
0x1C13							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RO	No	0~2	1	-
0	Index of assigned TxPDO 1	UINT	RW	No	0x1A00 ~ 0x1A03	0x1A01	-

0x1C32 Sync Manager 2 Synchronization

Index		Sync Manager 2 Synchronization					
0x1C32							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RO	No	-	10	-
1	Synchronization modes	UINT	Read Only	No	-	0: Free-Run	-
2	Cycle time	UDINT	RO	No	-	Time between Sync0 events [ns]	-
3	Shift time	UDINT	RO	No	-	125000[ns]	-
4	Synchronization mode Supported	UINT	Read Only	No	-	bit0=1: Free-Run supported bit4:2=001: DC Sync0 supported bit6:5=00: No Output Shift supported	-
5	Minimum cycle time	UDINT	RO	No	-	62500[ns]	-
6	Calc. and copy time	UDINT	RO	No	-	62500[ns]	-
9	Delay time	UDINT	RO	No	-	0[ns]	-
10	Sync0 time	UDINT	RO	No	-	Same as 1C32:02	-
11	Cycle exceeded counter	UDINT	RO	No	-	0	
13	Shift too short counter	UDINT	RO	No	-	-	
32	Sync error	BOOL	RO	No	-	-	

0x1C33 Sync Manager 3 Synchronization

Index		Sync Manager 3 Synchronization					
0x1C33							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Number of entries	USINT	RO	No	-	10	-
1	Synchronization modes	UINT	Read Only	No	-	same as 0x1C32:01	-
2	Cycle time	UDINT	Read	No	-	same as 0x1C32:02	-
3	Shift time	UDINT	RO	No	-	125000*n[ns] (n=1,2,3,...) Range: 0 to (Sync0 cycle time - 125000)	-
4	Synchronization mode Supported	UINT	Read Only	No	-	bit0 = 1: Free-Run supported bit 4:2 = 001: DC Sync0 supported bit 6:5 = 01 : Input Shift with local timer supported	-
5	Minimum cycle time	UDINT	RO	No	-	same as 0x1C32:05	-
6	Calculation and copy time	UDINT	RO	No	-	62500[ns]	-
9	Delay time	UDINT	RO	No	-	0[ns]	-
10	Sync0 time	UDINT	RO	No	-	Same as 1C32:10	-
11	Cycle exceeded counter	UDINT	RO	No	-	-	
13	Shift too short counter	UDINT	RO	No	-	-	
32	Sync error	BOOL	RO	No	-	-	

8.4 Manufacturer-Specific Objects

Index		Motor ID					
0x2000							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Motor ID	UINT	RW	No	0-999	0	-
Index		Main Voltage Type					
0x2001							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Main Voltage Type	UINT	RW	No	0-1	0	-
Index		U Current Offset					
0x2002							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	U Current Offset	INT	RW	No	-500-500	0	-
Index		V Current Offset					
0x2003							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	V Current Offset	INT	RW	No	-500-500	0	-
Index		CCW Torque Limit					
0x2004							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	CCW Torque Limit	UINT	RW	No	1-300	300	-
Index		CW Torque Limit					
0x2005							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	CW Torque Limit	UINT	RW	No	1-300	300	-
Index		Hidden Parameter No					
0x2006							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Hidden Parameter No	UINT	RW	No	0-19	0	-
Index		Hidden Parameter Value					
0x2007							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Hidden Parameter Value	UINT	RW	No	0-9999	0	-

Index		Start Menu Display No					
0x2008							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Start Menu Display No	UINT	RW	No	0-19	1	-
Index		Motor Inertia					
0x2009							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Motor Inertia	UINT	RW	No	1-9999	Motor ID	-
Index		Torque Constant					
0x200A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Torque Constant	UINT	RW	No	1-9999	Motor ID	-
Index		Phase Inductance					
0x200B							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Phase Inductance	UINT	RW	No	100-9999	Motor ID	-
Index		Phase Resistance					
0x200C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Phase Resistance	UINT	RW	No	1-9999	Motor ID	-
Index		Rated Current					
0x200D							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Rated Current	UINT	RW	No	1-9999	Motor ID	-
Index		Max Speed					
0x200E							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Max Speed	UINT	RW	No	1-6000	Motor ID	-
Index		Rated Speed					
0x200F							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Rated Speed	UINT	RW	No	1-6000	Motor ID	-
Index		Pole Number					
0x2010							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Pole Number	UINT	RW	No	2-16	Motor ID	-

Index		Gain Group					
0x2011							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Gain Group	UINT	RW	No	0-9	0	-
Index		Encoder Type					
0x2012							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Encoder Type	UINT	RW	No	0-5	0	-
Index		Encoder Pulse					
0x2013							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Encoder Pulse	UINT	RW	No	1-99999	32768	-
Index		Inertia Ratio					
0x2014							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Inertia Ratio	UINT	RW	No	1-50	1	-
Index		Speed P Gain1					
0x201A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed P Gain1	UINT	RW	No	1-5000	500	-
Index		Speed P Gain2					
0x201B							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed P Gain2	UINT	RW	No	1-5000	500	-
Index		Speed I TimeConstant1					
0x201C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed I TimeConstant1	UINT	RW	No	1-1000	20	-
Index		Speed I TimeConstant2					
0x201D							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed I TimeConstant2	UINT	RW	No	1-1000	20	-
Index		Speed Cmd Filter					
0x201E							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed Command Filter	UINT	RW	No	10-9999	50	

Index		Torque Command Filter					
0x201F							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Torque Command Filter	UINT	RW	No	10-9999	50	-
Index		Communication Type					
0x2025							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Communication Type	UINT	RW	No	0-1	0	-
Index		Analog Speed Command					
0x2028							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Analog Speed Command	UINT	RW	No	1-6000	3000	-
Index		Analog Speed Offset					
0x2029							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Analog Speed Offset	INT	RW	No	-1000-1000	0	-
Index		Zero Speed Clamp					
0x202A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Zero Speed Clamp	UINT	RW	No	0-1	0	-
Index		Zero Speed Clamp Voltage					
0x202B							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Zero Speed Clamp Voltage	UINT	RW	No	1-1000	1	-
Index		Speed Override					
0x202C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed Override	UINT	RW	No	0-1	0	-
Index		Analog Torque Command					
0x202D							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Analog Torque Command	UINT	RW	No	1-300	100	-
Index		Analog Torque Offset					
0x202E							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Analog Torque Offset	INT	RW	No	-1000-1000	0	-

Index		Zero Torque Clamp Mode					
0x202F							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Zero Torque Clamp Mode	UINT	RW	No	0-1	0	-
Index		Zero Torque Clamp Voltage					
0x2030							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Zero Torque Clamp Voltage	UINT	RW	No	1-1000	1	-
Index		Monitor Type1					
0x2031							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Type1	UINT	RW	No	0-5	1	-
Index		Monitor Mode1					
0x2100							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Mode1	UINT	RW	No	0-1	1	-
Index		Monitor Scale1					
0x2101							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Scale1	UINT	RW	No	1-500	10	-
Index		Monitor Offset1					
0x2102							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Offset1	INT	RW	No	-1000-1000	0	-
Index		Monitor Type2					
0x2103							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Type2	UINT	RW	No	0-5	1	-
Index		Monitor Mode2					
0x2104							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Mode2	UINT	RW	No	0-1	1	-
Index		Monitor Scale2					
0x2105							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Scale2	UINT	RW	No	1-500	10	-

Index		Monitor Offset2					
0x2106							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Monitor Offset2	INT	RW	No	-1000-1000	0	-
Index		Torque Direction					
0x2107							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Torque Direction	UINT	RW	No	0-1	0	-
Index		Register Brake Time					
0x2108							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Reg Brake Time	UINT	RW	No	100-1000	500	-
Index		SPD/TRQ Offset Mode					
0x2109							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	SPD/TRQ Offset Mode	UINT	RW	No	0-3	0	-
Index		InPosition Value					
0x210A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	InPosition Value	UINT	RW	No	1-9999	10	-
Index		Follow Error Range					
0x210B							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Follow Error Range	UINT	RW	No	1-9999	900	-
Index		Zero Speed Range					
0x210C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Zero Speed Range	UINT	RW	No	1-6000	10	-
Index		InSpeed Value					
0x210D							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	InSpeed Value	UINT	RW	No	1-500	100	-
Index		Brake Speed					
0x210E							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Brake Speed	UINT	RW	No	1-6000	50	-

Index		Brake Delay Time					
0x210F							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Brake Delay Time	UINT	RW	No	0-1000	10	-
Index		Power Fail Mode					
0x2110							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Power Fail Mode	UINT	RW	No	0-1	1	-
Index		Ready Mode					
0x2111							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Ready Mode	UINT	RW	No	0-1	0	-
Index		Pulse Clear Type					
0x2112							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Pulse Clear Type	UINT	RW	No	0-1	1	-
Index		Pulse Output Numerator					
0x2113							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Pulse Output Numerator	UINT	RW	No	1-9999	1	-
Index		Pulse Output Denominator					
0x2114							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Pulse Output Denominator	UINT	RW	No	1-9999	1	-
Index		Encoder A/B Lead					
0x2115							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Encoder A/B Lead	UINT	RW	No	0-1	1	-
Index		PWM On Delay Time					
0x2116							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	PWM On Delay Time	UINT	RW	No	0-5000	25	-
Index		Speed Direction Mode					
0x2117							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed Direction Mode	UINT	RW	No	0-1	0	-

Index		Output Logic Set					
0x2118							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Output Logic Set	UINT	RW	No	0-111	001	-
Index		PWM Off Delay Time					
0x2119							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	PWM Off Delay Time	UINT	RW	No	0-5000	0	-
Index		Input Logic Set Low					
0x211A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Input Logic Set Low	UINT	RW	No	0-1111	001	-
Index		Input Logic Set High					
0x211B							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Input Logic Set High	UINT	RW	No	0-1111	000	-
Index		Zero Speed Gain High					
0x211C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Zero Speed Gain High	UINT	RW	No	0-1	1	-
Index		Dynamic Brake Control Mode					
0x211D							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Dynamic Brake Control Mode	UINT	RW	No	0-3	0	-
Index		Operation Mode					
0x211E							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Operation Mode	UINT	RW	No	0-5	1	-
Index		Speed Command1					
0x211F							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed Command1	INT	RW	No	-6000-6000	100	-
Index		Speed Command2					
0x2120							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed Command2	INT	RW	No	-6000-6000	500	-

Index		Speed Command3					
0x2121							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Speed Command3	INT	RW	No	-6000-6000	1000	-
Index		Dynamic Brake Delay Time					
0x2122							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Dynamic Brake Delay Time	UINT	RW	No	0-2000	1	-
Index		Z Phase Search Speed					
0x2124							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Z Phase Search Speed	UINT	RW	No	1-300	10	-
Index		Manual Position Speed					
0x2125							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Manual Position Speed	UINT	RW	No	1-6000	500	-
Index		Acceleration Time					
0x2126							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Acceleration Time	UINT	RW	No	0-9999	0	-
Index		Deceleration Time					
0x2127							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Deceleration Time	UINT	RW	No	0-9999	0	-
Index		S-Curved Mode					
0x2128							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	S-Curved Mode	UINT	RW	No	0-1	0	-
Index		Test Run Speed0					
0x2129							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Speed0	INT	RW	No	-6000-6000	100	-
Index		Test Run Speed1					
0x212A							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Speed1	INT	RW	No	-6000-6000	100	-

Index		Test Run Speed2					
0x212B							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Speed2	INT	RW	No	-6000-6000	-500	-
Index		Test Run Speed3					
0x212C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Speed3	INT	RW	No	-6000-6000	1000	-
Index		Test Run Time0					
0x212D							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Time0	UINT	RW	No	1-3600	5	-
Index		Test Run Time1					
0x212E							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Time1	UINT	RW	No	1-3600	5	-
Index		Test Run Time2					
0x212F							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Time1=2	UINT	RW	No	1-3600	5	-
Index		Test Run Time3					
0x2130							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Test Run Time3	UINT	RW	No	1-3600	5	-
Index		InSpeed Type					
0x2131							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	InSpeed Type	UINT	RW	No	0-9999	1	-
Index		Input Pulse Type					
0x2200							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Input Pulse Type	UINT	RW	No	0-9999	0	-
Index		Electric Gear Numerator0					
0x2201							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Numerator0	UINT	RW	No	1-9999	1000	-

Index		Electric Gear Denominator0					
0x2202							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Denominator0	UINT	RW	No	1-9999	1000	-
Index		Electric Gear Numerator1					
0x2203							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Numerator0=1	UINT	RW	No	1-9999	2000	-
Index		Electric Gear Denominator1					
0x2204							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Denominator1	UINT	RW	No	1-9999	1000	-
Index		Electric Gear Numerator2					
0x2205							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Numerator2	UINT	RW	No	1-9999	3000	-
Index		Electric Gear Denominator2					
0x2206							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Denominator2	UINT	RW	No	1-9999	1000	-
Index		Electric Gear Numerator3					
0x2207							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Numerator3	UINT	RW	No	1-9999	4000	-
Index		Electric Gear Denominator3					
0x2208							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Electric Gear Denominator3	UINT	RW	No	1-9999	1000	-
Index		Pulse Input Direction					
0x220C							
Sub Index	Name	Data Type	Access	PDO Mapping	Setting Range	Initial Value	Unit
0	Pulse Input Direction	UINT	RW	No	0-1	0	-

Appendix - 1

Noise Countermeasures

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A-1.1 Types of Noise

Types of noise include noise generated by the servo drive's power section high-speed switching devices (IGBTs), repetitive switching operations of relay contacts, and inductive loads from internal electronic components. It also includes noise caused by external static electricity, surges, voltage fluctuations, and electromagnetic waves from other devices. Noise can cause product malfunctions and performance degradation, so please ensure proper grounding and wiring.

1) Internal Radiated Noise in the Servo Drive

- IGBT high-speed switching noise
- Electromagnetic inductive noise from transformers
- Relay switching noise
- Pulse noise from DSP clocks, etc.

2) External influence noise

- Power line radiated noise
- Use of electronic contactors, electronic brakes, and relays
- Use of high-inductance electrical and electromagnetic devices

A-1.2 Noise Countermeasures

1) Servo Drive Radiated Noise Countermeasures

- Install and route noise-sensitive external peripherals as far away from the servo drive as possible.
- Do not run the servo drive's power cable, encoder cable, I/O cable, etc., in parallel or bundled wiring.
- Use shielded cables for the servo drive's I/O cable (CN1) and encoder cables (CN2, CN3), or route them through separate metal conduits.

2) External Noise Countermeasures

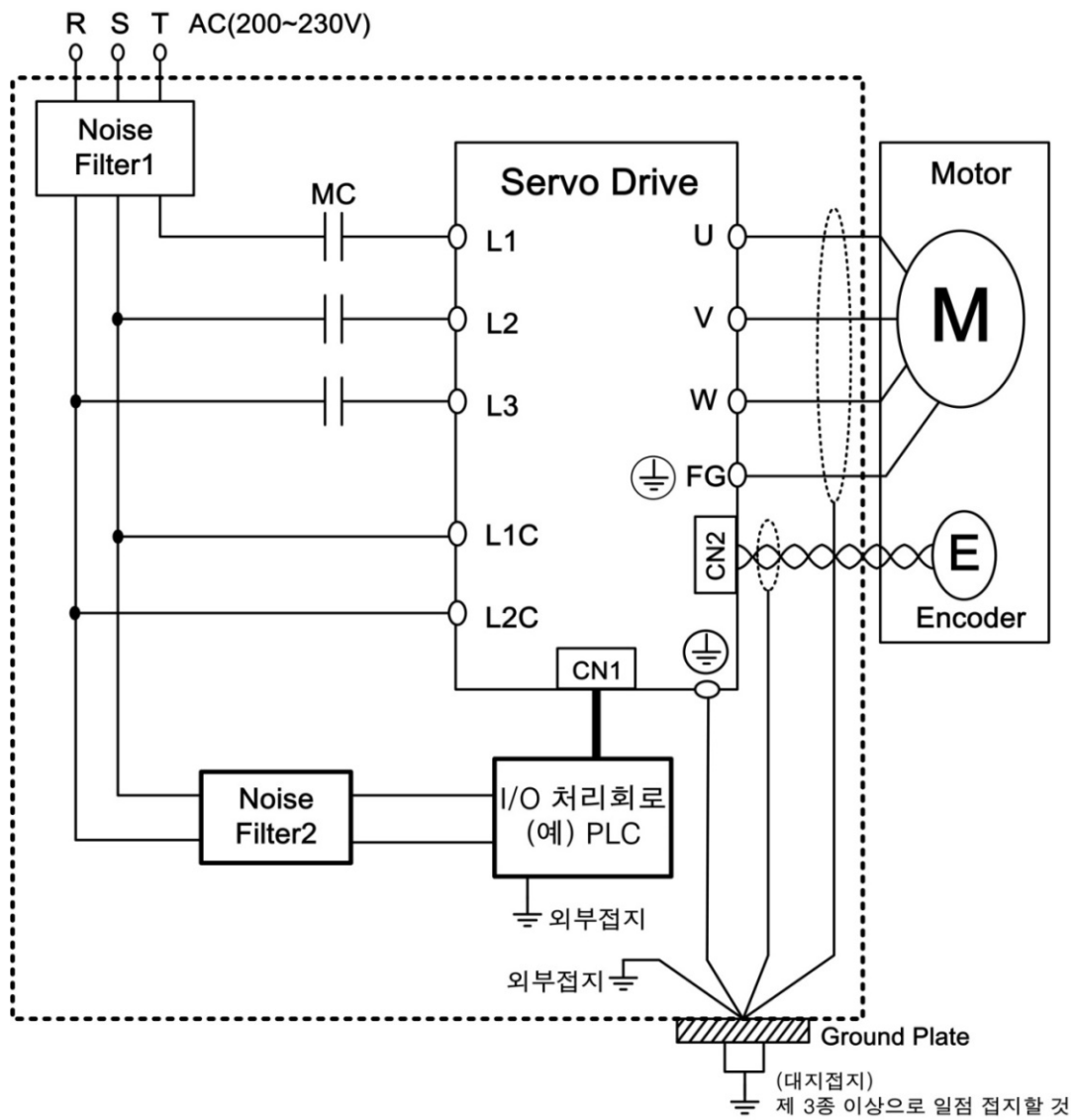
- Be sure to install surge killers on peripheral devices that are sources of noise.
- Ground the I/O cable (CN1) and encoder cables (CN2, CN3) using the internal clamp metal fittings.
- Attach line filters to signal lines.
- If a closed-loop circuit is formed by the encoder wiring and signal line ground wires, leakage current may flow, causing device malfunction. Disconnecting the ground wires can prevent this

malfunction.

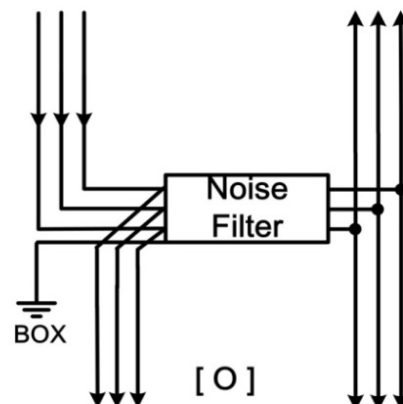
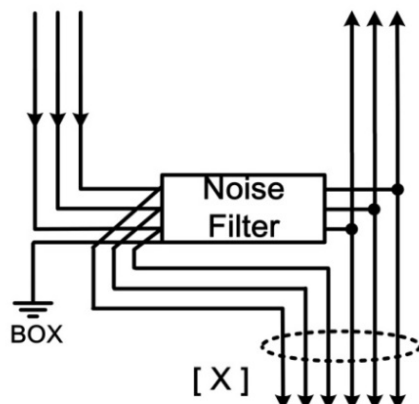
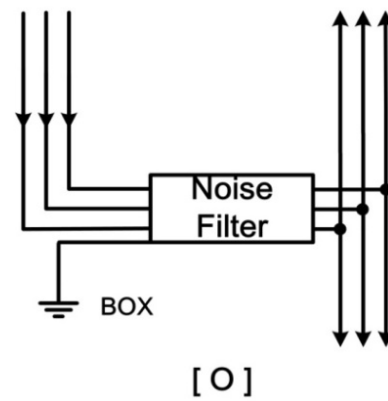
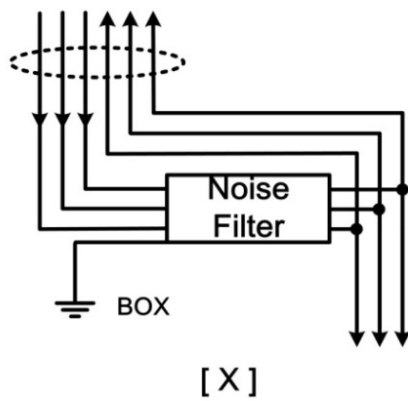
A-1.3 Wiring for Noise Reduction

1) Grounding-Related Wiring Example

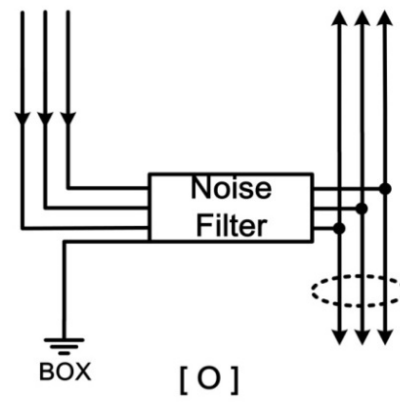
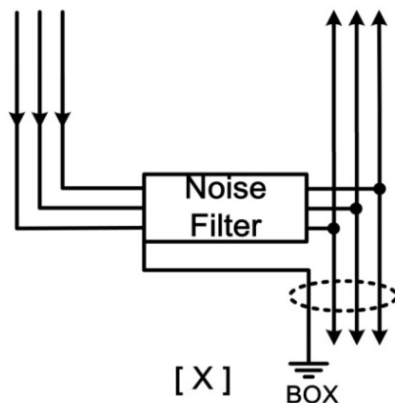
- Use a thick wire of 3.5 mm^2 or larger for the ground wire.



2) Noise Filter Wiring

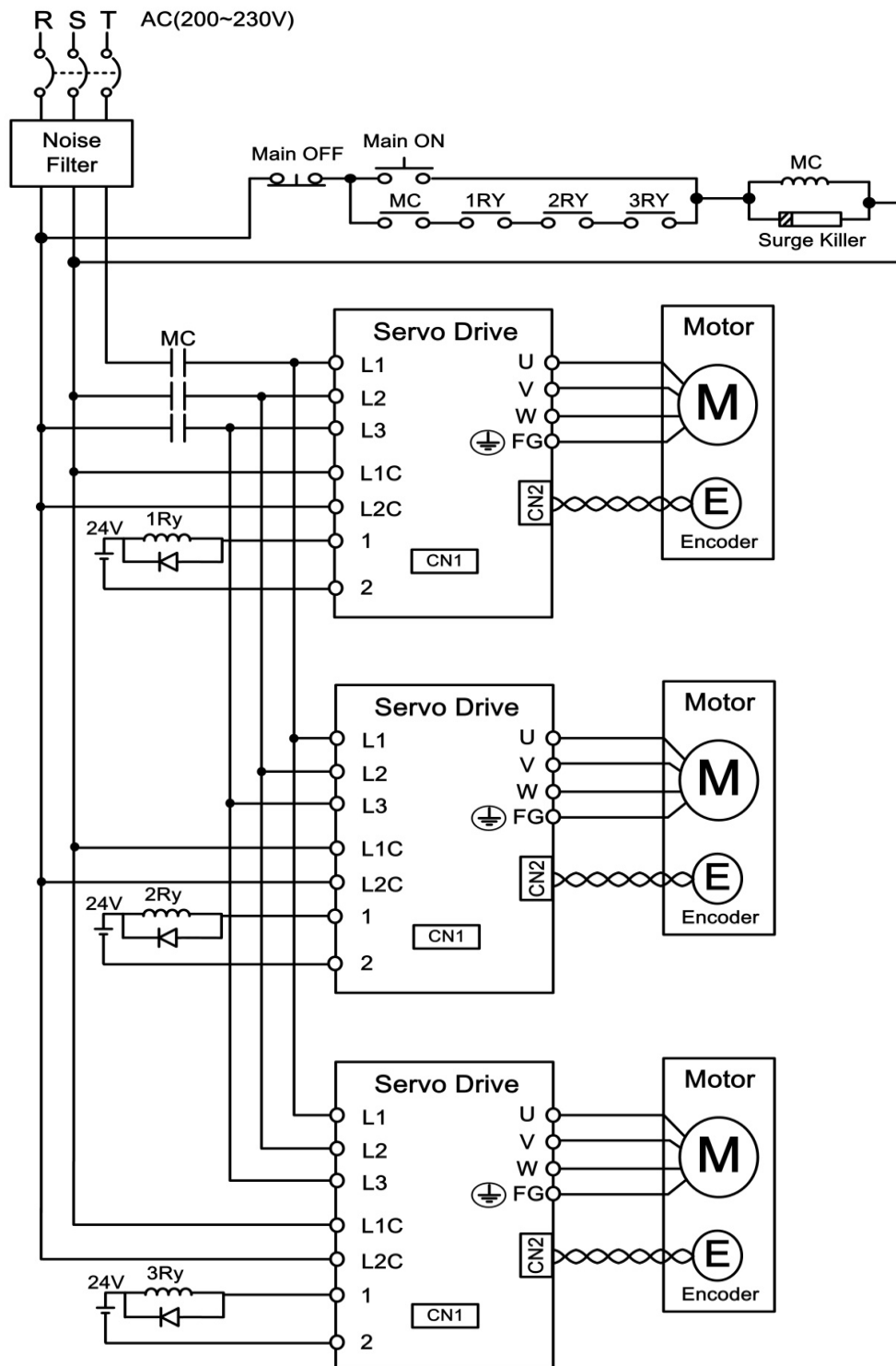


[입력선과 출력선을 같이 묶거나 같은 DUCT에 넣으면 안됨]



[NF의 접지선과 출력선을 같이 묶거나 같은 DUCT에 넣으면 안됨]

3) Wiring for Connecting Multiple Servo Drives



Appendix - 2

Servo Parameter List

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A-2.1 Operation Status Display Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
dP000	Current Operating Status	-	-	Displays the current operating status. Servo On: dr-on, Servo Off: dr-of, Alarm Status: Alarm Number
	Current State	-	-	
dP001	Current Operating Speed	r/min	0	Displays the current operating speed.
	Current Speed	-10000	10000	
dP002	Current Command Speed	r/min	0	Displays the current command speed.
	Command Speed	-10000	10000	
dP003	Not Used			
	Not Used			
dP004	Not Used			
	Not Used			
dP005	Not Used			
	Not Used			
dP006	Not Used			
	Not Used			
dP007	Current Command Torque	%	0	Displays the current command torque during torque control operation.
	Command Torque	-300.00	300.00	
dP008	Torque Limit	%	0	Displays the torque limit setting value.
	Torque Limit	-300.00	300.00	
dP009	Current Load Rate	%	0	Displays the current load rate relative to the rated value.
	Current Load	0.00	300.00	
dP010	Average Load Factor	%	0	Displays the average load factor over 5 seconds relative to the rated value.
	Average Load	0.00	300.00	
dP011	Instantaneous Maximum Load Factor	%	0	Indicates the maximum instantaneous load factor relative to the rated value.
	Maximum Load	-300.00	300.00	
dP012	DC Link Voltage	V	0.0	Displays the DC Link capacitor voltage of the current main power supply.
	DC Link Voltage	0.0	500.0	
dP013	CN1 Contact Status	-	-	Displays the hardware connection status of the CN I/O contact . Output contacts display the output status.
	I/O Display	-	-	

dP014	Regenerative Braking Operation Time	ms	0	Displays the regenerative braking operation time.
	Reg. Break Time	0	3000	
dP015	Input/Output Contact Status	-	-	Displays the results of the recognized input/output contact setting status, I/O contact setting status recognized from the CN I/O contact.
	I/O State Display	-	-	

dP016	Not Used	-	-	
	Not Used	-	-	
dP017	Not Used	-	-	
	Not Used	-	-	
dP018	Not Used	-	-	
	Not Used	-	-	
dP019	Software Version	-	-	Displays the software version number.
	Software Version	-	-	

A-2.2 System Variable Settings Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
*PD-200	Motor ID	-	-	Sets the motor ID.
	MOTOR ID	-	-	
PD-201	Main Voltage Rating Setting		0	Sets the rated voltage of the main power supply. 0: AC220V, 1: AC440V
	Main Voltage Type	0	1	
PD-202	U-phase Current Offset Setting	mA	0	Displays the U-phase current offset. -Modification is possible when Motor ID [PD200] is "0".
	U Current Offset	-500	500	
PD-203	Phase V Current Offset Display	mA	0	Displays the V-phase current offset. -Modification is possible when Motor ID [PD200] is "0".
	V Current Offset	-500	500	
PD-204	Forward Torque Limit	%	300	Sets the torque limit value during clockwise rotation.
	CCW Torque Limit	1	300	

PD-205	Reverse rotation torque limit	%	300	Sets the torque limit value during reverse rotation.
	CW TRQ Limit	1	300	
PD-206	Hidden Menu Number	-	0	Set the variable number for the hidden menu [HD900~HD919]. .
	Hidden Parm No	0	19	
PD-207	Hidden Menu Value	-	0	Sets the variable value for Hidden Menu Number [PD-207].
	Hidden Parm Val	0	9999	
PD-208	Display Menu at Startup	-	0	Sets the drive's operation status display menu. -Displays the set number [DP000~DP019] when servo power is ON
	Start Menu No.	0	19	
PD-209	Inertia	gf.cm.s ²	ID	Sets the motor inertia moment. -Modification is possible when the motor ID [Pd200] is "0".
	Inertia	0.01	99.99	

PD-210	Motor Torque Constant	Kgf-cm/A	ID	Sets the motor torque constant. -Modification is possible when the motor ID [Pd200] is "0".
	Torque Const.	0.01	99.99	
PD-211	Motor Stator Inductance	mH	ID	Sets the motor shunt inductance. -Modification is possible when Motor ID [Pd200] is "0".
	Phase Ls	0.1	9.999	
PD-212	Motor Resistance	ohm	ID	Sets the motor phase resistance. -Modification is possible when Motor ID [Pd200] is "0".
	Phase Rs	0.001	9.999	
PD-213	Motor Rated Current	A	ID	Sets the motor rated current. -Modification is possible when Motor ID [Pd200] is "0".
	Rated Is	0.01	99.99	
PD-214	Motor Maximum Speed	r/min	ID	Sets the motor maximum speed. -Modification is possible when Motor ID [Pd200] is "0".
	Max Speed	1	6000	
PD-215	Motor Rated Speed	r/min	ID	Sets the motor rated speed. -Modification is possible when Motor ID [Pd200] is "0".
	Rated Speed	1	6000	
PD-216	Motor Pole Count	Pole	ID	Sets the motor pole count. -Modification is possible when the motor ID [Pd200] is "0".
	Pole Number	2	16	
PD-217	Motor Gain Group	-	ID	Sets the motor gain group. -Modification is possible when the motor ID [Pd200] is "0".
	Gain Group	0	9	
**PD-218	Encoder Type	-	0	Sets the encoder type.
	Encoder Type	0	5	
**PD-219	Encoder Pulse Count	ppr	32768	Set the number of pulses per revolution of the applied encoder's electrical angle. -Modification is possible when the Motor ID [Pd200] is "0".
	Encoder Pulse	1	32768	

A-2.3 Control Variable Setting Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
PC-300	System Inertia Ratio	-	1.0	Sets the load's inertia ratio.
	Inertia Ratio	1.0	50.0	
PC-301	Position P-Gain 1	Hz	50	Sets Position Proportional Gain 1.
	Position Proportional Gain 1	1	500	

PC-302	Position P-Gain 2	Hz	70	Sets Position Proportional Gain 2.
	Position Proportional Gain 2	1	500	
PC-303	Position FF Rate	%	0	Sets the position feedforward ratio.
	Feedforward Ratio	0	100	
PC-304	Position FF TC	ms	50	Sets the position feedforward correction factor.
	Feedforward correction factor	0	9999	

PC-305	Not Used			Not Used
	Not Used			
PC-306	Speed Proportional Gain 1	rad/s	400	Sets Speed Proportional Gain 1
	Speed P Gain1	1	5000	
PC-307	Speed Proportional Gain 2	rad/s	400	Set Speed Proportional Gain 2
	Speed P Gain2	1	5000	
PC-308	Speed Integral Constant 1	ms	25	Set Speed Integral Time Constant 1.
	Speed I TC1	1	1000	
PC-309	Speed Integral Time Constant 2	ms	25	Set Speed Integral Time Constant 2
	Speed I TC2	0.0	1000	
PC-310	Speed Command Correction Factor 2	ms	0.50	Sets the speed command filter.
	Speed CMD TC	0.1	99.99	
PC-311	Speed Feedback Correction Factor	ms	0.50	Sets the speed feedback filter.
	Speed F/B FLT	0.1	99.99	
PC-312	Not used			Not used
	Not Used			
PC-313	Torque Command Correction Value	ms	0.50	Sets the torque command filter
	TRQ CMD TC	0.1	99.99	
PC-314	Not used			Not Used
	Not Used			
PC-315	Not Used			Not Used
	Not Used			
PC-316	Not Used			Not Used
	Not Used			

PC-317	Communication Method	-	0	Sets the communication method for the servo drive. -EtherCAT: 0, Analog: 1
	Communication Type		1	
PC-318	Not Used			Not Used
	Not Used			
PC-319	Not Used			Not Used
	Not Used			

A-2.4 Analog Control Variable Setting Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
*PA-400	±10[V] Analog Speed Command	r/min	2000	Sets ±10V analog speed input.
	±10[V] Analog Speed	1	6000	
PA-401	Speed Command Offset	mV	0	Sets the offset for the analog speed command.
	Speed Offset	-1000	1000	
PA-402	SPD Clamp Mode	-	0	Sets the clamp function for analog zero speed commands.
	Speed Clamp Mode Setting	0	1	
PA-403	Clamp Voltage	mV	1	Sets the analog zero-speed clamp operating range. Applies when [PA-402] = 1
	Clamp Voltage Setting	1	1000	
*PA-404	Speed Override	-	0	Enables the Speed Override driving function
	Speed Override	0	1	
*PA-405	±10[V] Analog Torque Command	%	100	Sets the analog torque command at ±10V
	±10[V] Analog Torque	1	300	
PA-406	Torque Command Offset	mV	0	Sets the offset for the analog torque command.
	Torque Offset	-1000	1000	
PA-407	Torque Clamp Mode	-	0	Sets the clamp function for analog zero torque commands.
	TRQ Clamp Mode	0	1	
PA-408	Torque Clamp Voltage	mV	1	Sets the analog zero torque command clamp voltage. Applies when [PA-407] = 1
	Torque Clamp Voltage	1	1000	
PA-409	Analog Output Format 1	-	1	Sets the format for Monitor Analog Output 1.
	Monitor Type 1	0	5	
PA-410	Analog Output Mode 1	-	0	Sets the mode for Analog Output 1 for the monitor.
	Monitor Mode 1	0	1	
PA-411	Analog Output Scale 1	-	1.0	Sets the gain for Analog Output 1 for monitors. Set based on a 5V voltage reference.
	Monitor Scale 1	0.1	50.0	
PA-412	Analog Output Offset 1	mV	0	Sets the offset for Monitor Analog Output 1.
	Monitor Offset1	-1000	1000	
PA-413	Analog Output Format 2	-	1	Sets the format for Monitor Analog Output 2.
	Monitor Type 2	0	5	
PA-414	Analog Output Mode 2	-	0	Sets the mode for Analog Output 2 for monitors
	Monitor Mode 2	0	1	
PA-415	Analog Output Scale 2	-	1.0	Sets the gain for Analog Output 2 for monitors. Set based on the 5V voltage.
	Monitor Scale 2	0.1	50.0	

PA-416	Analog Output Offset 2	mV	0	Sets the offset for Monitor Analog Output 2.
	Monitor Offset 2	-1000	1000	

PA-417	Torque Command Direction Setting	-	0	Selects the motor operating direction relative to the torque command voltage during torque control operation.
	Torque CMD DIR	0	1	
PA-418	Regenerative Braking Operation Time	ms	500	Sets the maximum operation time for the regenerative braking resistor.
	Reg. Brake Time	100	1000	
PA-419	Speed/Torque Offset	-	0	[CH-812][CH-813] Analog offset function is applied.
	SPD/TRQ Offset	0	3	

A-2.5 Input/Output Variable Settings Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
PF-500	Position Completion Range	10Pulse	10	Sets the output range for the position operation completion signal.
	In Position	1	9999	
PF-501	Tracking Error Error Pulse	100Pulse	900	Sets the output range for the positional tracking error signal.
	Follow Error	1	9999	
PF-502	Zero Speed Reach Range	r/min	10	Sets the zero speed signal output range.
	Zero Speed Range	0.0	6000	
PF-503	Speed Operation Completion Range	r/min	100	Sets the speed reach completion signal output range.
	InSpeed Value	0.0	500	
PF-504	Brake Operation Speed	r/min	50	Sets the brake operation signal output speed.
	Brake Speed	0.0	6000	
PF-505	Brake Operation Time	ms	10	Sets the brake operation signal output delay time.
	Brake Time	0	1000	
PF-506	Main Power Abnormal Mode	-	1	Sets the operation reset mode for main power failure errors.
	Power Fail Mode	0	1	
PF-507	READY Contact Mode	-	0	Selects the output contact for Ready/Break.
	Ready Mode	0	1	

PF-508	Not Used			Not Used
	Not Used			
**PF-509	Encoder Subdivision Molecule	-	1	Sets the encoder output pulse division ratio
	Pulse Output Number	1	9999	
**PF-510	Encoder Divider Denominator	-	1	Sets the denominator of the encoder output pulse division ratio.
	Pulse Output Den	1	9999	
**PF-511	Encoder A/B Phase Out	-	0	Sets the direction of encoder output pulses for A/B phase
	A/B Phase Lead	0	1	

PF-512	Not Used			Not Used
	Not Used			
PF-513	Speed and Driving Direction Settings	-	0	Set the operation method for the driving direction change switch
	DIR Select Mode	0	1	
*PF-514	Output Contact Logic Setting	-	001	Sets the logic for the output contact.
	Output Logic	000	111	
PF-515	PWM OFF Delay Time	ms	0	Sets the PWM output OFF delay time when the servo is OFF.
	PWM OFF Delay	10	1000	
*PF-516	Input Logic Setting 0~3	-	1100	Sets the logic state for input contacts 0 to 3.
	Input Logic 0~3	0000	1111	
*PF-517	Input Logic Settings 4~7	-	0000	Sets the logic state for input contacts 4 to 7.
	Input Logic 4~7	0000	1111	
PF-518	Zero Speed Gain Ratio	%	50	Sets the steady-state gain ratio.
	ZSPD Gain Rate	1	100	
*PF-519	Gain Switching Mode	-	0	When the setting is 0, it sets the gain mode.
	Gain Conv. Mode	0	1	

A-2.6 Speed Operation Variable Setting Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
*PU-600	Operation Mode Selection	-	1	Sets the operation mode.
	Operation Mode	0	5	
PU-601	Internal Speed Command 1	r/min	200	Speed command input contact SPEED1 / SPEED2 combination . *SPD1,2 contacts are used for speed limiting in torque drive mode. .
	Speed Command1	-6000	6000	
PU-602	Internal Speed Command 2	r/min	1000	
	Speed Command 2	-6000	6000	
PU-603	Internal Speed Command 3	r/min	3000	
	Speed Command 3	-6000	6000	
PU-604	Not used	-	-	
	Not Used	-	-	

PU-605	Not Used	-	-	Not used
	Not Used	-	-	
PU-606	Z-axis Search Speed Setting	r/min	10	Encoder Z-phase search operation speed setting. This is the setting value for operation of [CH-805].
	Z Phase Search Spd	1	300	

PU-607	Manual Positioning Speed	r/min	500	Sets the operating speed during manual position operation. -This is the setting value for operation of [CH-806].
	Manual Pos Spd	1	6000	
PU-608	Speed Command Acceleration Time	ms	0	Sets the speed command acceleration time.
	Accel Time	0	9999	
PU-609	Speed Command Deceleration Time	ms	0	Sets the deceleration time for the speed command.
	Decel Time	0	9999	
*PU-610	S-curve driving	-	0	Sets the S-curve driving motion during speed control. 0 : Trapezoidal, 1 : S-Curve
	S-Curve Control	0	1	
PU-611	Jog Speed 0	r/min	100	Set speed to 0 during continuous test operation.
	Jog Speed 0	-6000	+6000	
PU-612	Jog Speed1	r/min	100	Set speed 1 during continuous test operation.
	JOG Speed1	-6000	+6000	
PU-613	Jog Speed 2	r/min	100	Set speed 2 during continuous test operation.
	Jog Speed 2	-6000	+6000	
PU-614	Jog Speed 3	r/min	100	Set speed 3 during continuous test operation.
	JOG Speed 3	-6000	+6000	
PU-615	Jog Time 0	sec	5	Set the operation time to 0 during continuous test operation.
	Jog Time0	1	3600	
PU-616	Jog Time1	sec	5	Set the jog time 1 during continuous test operation.
	JOG Time1	1	3600	
PU-617	Jog Time2	sec	5	Set the jog time 2 during continuous test operation.
	JOG Time2	1	3600	
PU-618	Jog Time3	sec	5	Set the jog time to 3 during continuous test operation.
	JOG Time3	1	3600	
PU-619	INSPEED Output Format	-	1	Sets the output type for the INSPD output contact. 0: Zero speed output, 1: Speed matching
	InSpeed Type	-0	1	

A-2.7 Position Operation Variable Setting Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
PP-700	Position Control Acceleration Time	-	1	Sets the acceleration time when using position control mode.
	POS ACC	1	1000	
PP-701	Electronic Gear Ratio Numerator 0		1000	Sets the electronic gear ratio 0 molecular term.
	Elec Gear Num0	1	9999	

PP-702	Electro Gear Ratio Denominator 0		1000	Sets the denominator term for the electronic gear ratio.
	Elec Gear Den0	1	9999	
PP-703	Electro Gear Ratio Numerator 1		1000	Sets the electronic gear ratio 1 molecular item.
	Elec Gear Num1	1	9999	
PP-704	Electro Gear Ratio Denominator 1		1000	Sets the denominator term for the first electromagnetic gear ratio.
	Elec Gear Den1	1	9999	
PP-705	Electro Gear Ratio Numerator 2		1000	Sets the electronic gear ratio 2 molecular value.
	Elec Gear Num2	1	9999	
PP-706	Electro Gear Ratio Numerator 2		1000	Sets the denominator term for the electronic gear ratio 2.
	Elec Gear Den2	1	9999	
PP-707	Electro Gear Ratio Numerator 3		1000	Sets the electronic gear ratio 3 molecular value.
	Elec Gear Num3	1	9999	
PP-708	Electro Gear Ratio Numerator 3		1000	Sets the denominator term for the electronic gear ratio 3.
	Elec Gear Den3	1	9999	
PP-709	Pulse Input Direction		0	Sets the direction of the pulse input.
	Pulse Input Direction	0	1	
PP-710	Not Used	-	-	Not Used
	Not Used	-	-	
PP-711	Not Used	-	-	Not Used
	Not Used	-	-	
PP-712	Not Used	-	-	Not Used
	Not Used	-	-	
PP-713	Not Used	-	-	Not Used
	Not Used	-	-	
PP-714	Not Used	-	-	Not Used
	Not Used	-	-	
PP-715	Not Used	-	-	Not Used
	Not Used	-	-	
PP-716	Absolute Singleton Sub	Pulse	0	Displays the single-turn lower 2 bytes of the absolute encoder.
	ABS Single Turn Low	0	65535	
PP-717	Absolute Single Turn High	Pulse	0	Displays the upper 2 bytes of the absolute encoder's singleton.
	ABS Single Turn High	0	65535	

PP-718	Absolute Multi-Turn Low	Pulse	0	Displays the lower 2 bytes of the absolute encoder's multi-turn value.
	ABS Multi-Turn Low	0	65535	
PP-719	Absolute Multi-Turn High	Pulse	0	Displays the upper 2 bytes of the absolute value encoder's multi-turn value.
	ABS Multi-Turn High	0	65535	

A-2.8 Operation Settings Menu

Menu		UNIT	UNIT	Description
Code	Name	MIN	MAX	
CH-800	Alarm Reset	-	-	Resets the alarm status.
	Alarm Reset	-	-	
CH-801	Alarm History Deletion	-	-	Clears the alarm history.
	Alarm History Clear	-	-	
CH-802	Manual Test Run	r/min	-	Perform manual test run.
	Manual Test Run			
CH-803	Continuous Test Run	r/min	-	Perform continuous test operation.
	Auto Test Run			
CH-804	Not used			Not Used
	Not Used			
CH-805	Z-phase search operation	turn	0.00	Perform a z-axis search operation at the speed set for [PU-606].
	Z Pos Search	0	9.99	
CH-806	Manual Position Operation	turn	-	Drives at the speed set in [PU-607] to the target position set in [PP-716]. at the speed set in [PU-607].
	Manual Position Run	-	-	
CH-807	Output Contact Forced On/Off	-	-	Temporarily sets the output contact forced On/Off. The contact set in [PF-514] is reflected.
	Output Test	-	-	
CH-808	Menu Reset	-	-	Resets the menu to factory default settings.
	Menu Data Initialization	-	-	
CH-809	Menu Locking	-	-	Applies the Menu Locking function.
	Menu Data Lock	-	-	
CH-810	Not Used	-	-	Not Used
	Not Used	-	-	
CH-811	Current Offset Compensation Function	-	-	Compensates for the current offset of the CT sensor.

	Current Offset Save	-	-	
CH-812	Speed Command Voltage Offset	mA	-	Automatically stores the Offset Data for the analog speed command voltage to the PA-401.
	Speed Cmd Offset	-	-	
CH-813	Torque Command Voltage Offset	mV	-	Automatically saves the Offset Data for the analog torque command voltage to the PA-406.
	Torque Cmd Offset		-	
CH-814	Instantaneous Maximum Load Rate	%	-	Displays the maximum instantaneous load factor.
	Maximum Load	-1000	1000	
CH-815	Tracking Position Pulse	Pulse	1	Displays the tracking position pulse.
	Feedback Pulse	0	99999	
CH-816	Not Used	-	-	Not Used
	Not Used	-	-	

CH-817	Not Used	-	-	Not Used
	Not Used	-	-	
CH-818	Not Used	-	-	Not used
	Not Used	-	-	
CH-819	Reset All Menu Values	-	-	Resets all menu data to its initial state.
	All Menu Data Init	-	-	

AC SERVO DRIVE

TURBO *SD Series with* Ether**CAT**  **Interface**
ver 1.0 (July 2015)

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